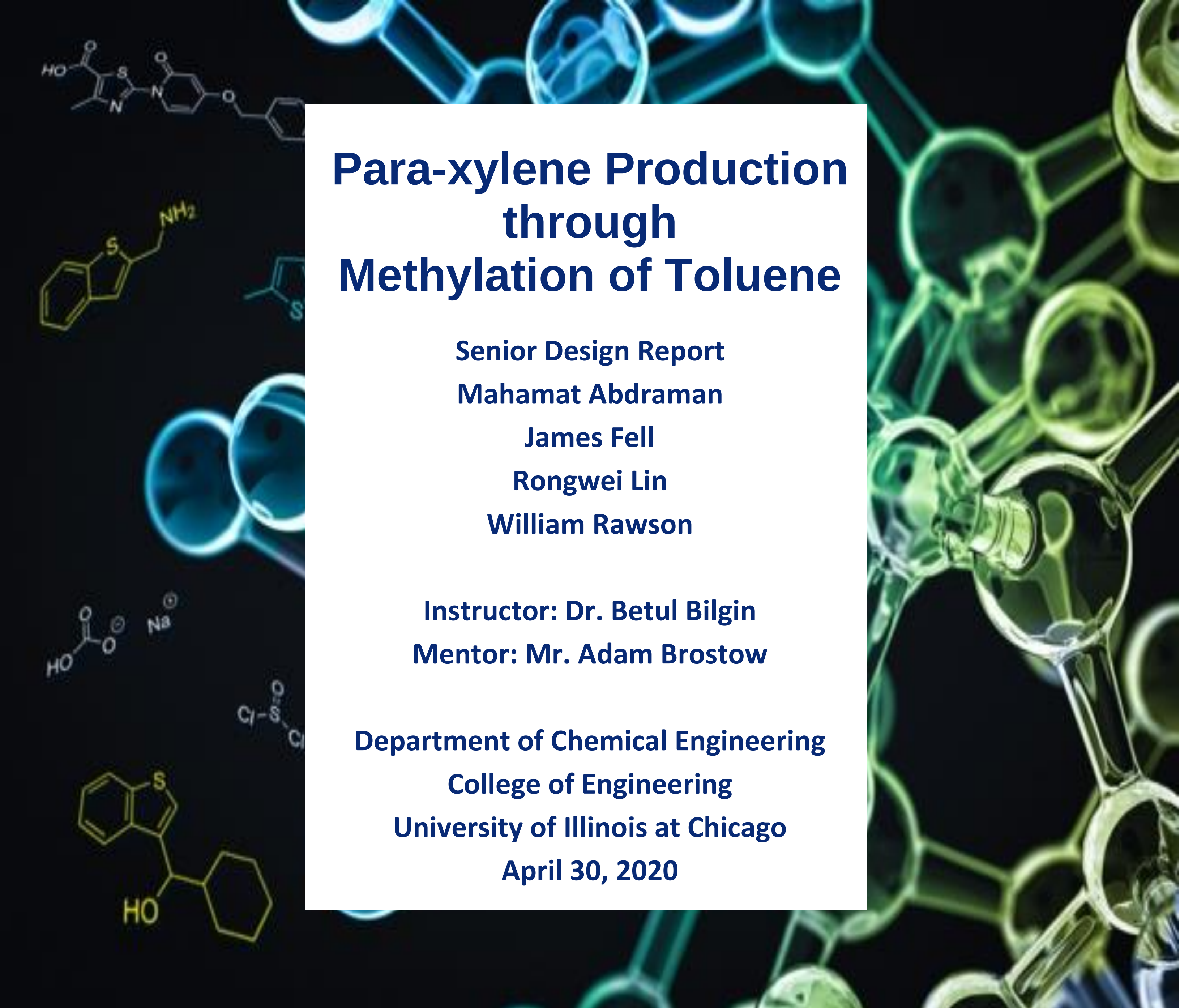

Para-xylene Production through Methylation of Toluene

Senior Design Report

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Executive Summary

Paraxylene is a commonly produced product in the petrochemical industry. This valuable product is used in the making of polyester fibers, films, and resins to satisfy the growing demand from emerging economies. The purpose of this project is to provide an innovative and novel design of a plant that will produce 99.5% pure paraxylene while maintaining the objectives of energy and cost efficiency. During the separation of paraxylene from a reformed naphtha feedstock, a large number of by-products are produced, namely toluene. Toluene has a relatively cheap market price and low demand. Implementing the methylation of toluene process to the plant will be able to convert toluene into paraxylene with a conversion of greater than 90%. The design will implement an existing technological model of toluene methylation while also incorporating new energy optimization ideas that can still produce para-xylene of optimal purity. The plant is expected to have a production capacity of 4.5 MMPTA.

The economic metrics justify the novel paraxylene production through toluene methylation compared to most conventional based and energy extensive processes. The process exhibits a high gross profit of $85.60 MM/year and a cash cost of production of $3,072.56 MM/year from a total revenue of $3,158.16 MM/year. High production capacity and relatively low energy consumption exhibited by the process makes toluene methylation more appealing to investors in developing countries with high demand. The highly increasing consumption of paraxylene-derived materials by emerging economies, such as China and India, indicate that global demand might grow even higher than the current rate of 7% per year. Therefore, the optimum production of paraxylene through toluene methylation is lucrative and will remain as such for the next few decades as more underdeveloped countries move towards economic prosperity.

Introduction

If you were to check the tag on the inside of your shirt, on your favorite bed sheets, or even the curtains draped over your living room windows, there is a high chance these fabrics contain polyester fiber. Even your favorite flick to see at the theatres is made possible by a polyester base coat on the movie film. Polyester fiber has a wide range of application, and in most cases used to improve desirable properties, such as wrinkle resistance in the textile industry [1]. With so many uses, polyester fiber is of high demand, and therefore the components used in its manufacture are as well.

Para-xylene is used as a building block in the manufacture of chemicals sought after in the plastic industry, such as terephthalic acid (TPA), purified terephthalic acid (PTA), and dimethyl-terephthalate (DMT) [1]. Of the three, polyester fiber is made possible by a para-xylene feedstock via the intermediate purified terephthalic acid (PTA) [1]. Generally, para-xylene is consumed as a polymer chain, making it a high value aromatic hydrocarbon. As a member of the BTX group (benzene, toluene, and xylenes), it is extracted from reformate, the product of catalytically reformed naphtha. Of the three possible isomers of dimethylbenzene found in the reformate, para-xylene has methyl groups at the 1,4 position and has the formula $C_6H_4(CH_3)_2$.

A typical para-xylene production plant has the capacity to produce both high purity para-xylene and benzene from a naphtha feedstock, with ortho-xylene as a co-product. At the plant, the naphtha feedstock is hydrated and sent to a separation unit that produces aromatic rich reformate. The reformate includes benzene, toluene, xylenes, and $C9^+$ aromatic compounds. With para-xylene being the desired product, it is purified and separated from the rest of the components. With such a big market for para-xylene, the separation and purification steps are designed by chemical engineers to produce high purity para-xylene. However, separation technology can be difficult and is constantly evolving because of the volatility and molecular weight are essentially equal for the xylenes. The separation processes are energy intensive and need to be optimized to produce the most cost-effective process. Furthermore, innovative technologies are able to produce bigger outputs of para-xylene, such as the methylation of toluene described in this report. This additional process can be implemented into a plant to meet future demands in the market, while also creating more revenue from a less valuable toluene by-product.

Economic Trends, Prices, and Demand for Para-xylene

Value Chain

Crude oil contains many types of fuels and hydrocarbon mixtures, like kerosene, natural gas, naphtha and tar for example. Naphtha is the initial feed stock in the production of paraxylene and must be separated from crude oil. When crude oil is distilled, the fraction that boils at 185-350 ^{o}F is considered the naphtha fraction. In the petrochemical industry, naphtha is a well-known mixture of heavy and light hydrocarbons. First, it is reformed, or reacted, to convert the hydrocarbons into aromatics and hydrogen before it is separated into its components. For the paraxylene process, it is most beneficial to reform the naphtha to produce the highest number of xylenes possible, which are benzenes with two methyl groups attached. Our desired product is paraxylene; a benzene ring with two methyl groups, one at position 1 and one at position 4.

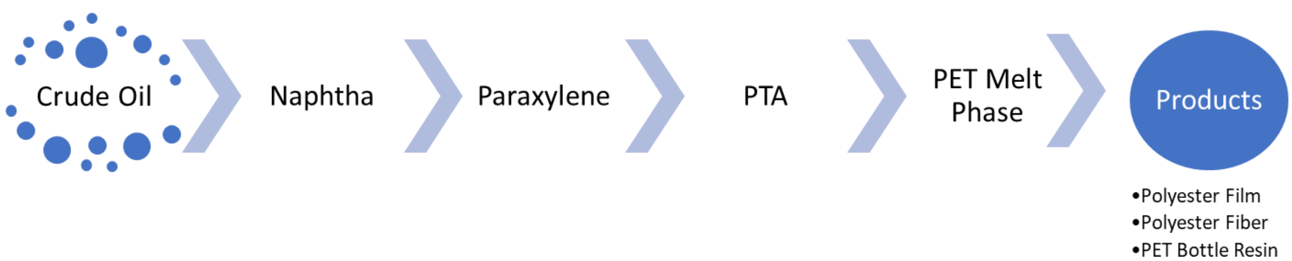


Figure 1: Value chain for the production and end use of paraxylene [1].

Paraxylene (PX) is the most desirable isomer of the xylene family. Around 98% of the paraxylene produced is turned into purified terephthalic acid (PTA) or dimethyl terephthalate (DMT) [1]. PTA and DMT are intermediates that can be turned into polyester fibers, films, or made into polyethylene terephthalate (PET) resin beads (See Figure 2). Other uses for PX include using it as a solvent, turning it into di-paraxylene, and using it in the manufacturing of herbicides. Paraxylene is an intermediate refined petrochemical, making it susceptible to pricing pressure from changes in the price of crude oil and the demand for polyester products (See Figure 1).

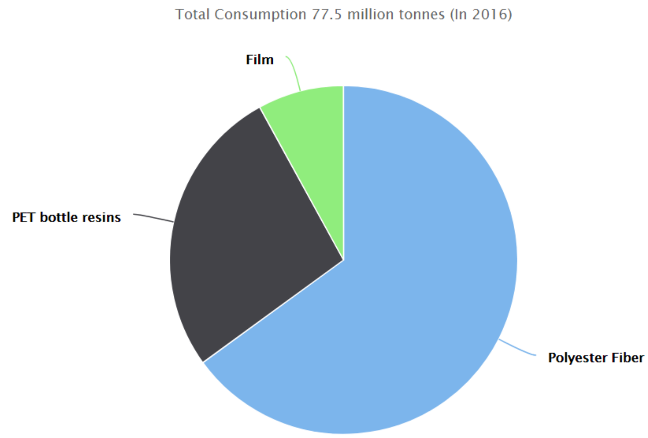


Figure 2: Demand of PTA per application. Application Demand in 2016: Polyester Fiber (65%), PET Bottle Resins (27%), Film (8%). Figure adapted from PlasticInsight.com [2].

Market Drivers

The market for PX has experienced strong growth. This is due to higher consumption rates of PTA, as the polyester manufacturing industry has been increasing output throughout the years. This increase in demand is due to the growth of the polyester fiber industry in Asia, which makes up two-thirds of the global demand (See Figure 3) [1]. In America and Europe, the production of polyester fibers has been declining as the textile industry has been shifting into Asia [1]. Though, in America and Europe, the consumption of PET has been quickly growing as more companies are preferring to switch from glass bottles to PET plastic bottles.

To support the rapidly growing textile industry in Asia, around 8.5 million metric tons per year of additional PX production capacity was commissioned in Northeast Asia in 2014 [3]. The commissioning of the new capacity caused an oversupply of PX and put downward pressure on the profitably of producing PX. This ultimately led to an oversupply of 10 million tons of PX and forced companies to decease operating rates to less than 80% [4,5]. Since then, steady growth in the consumption of PX and limited production of PX has greatly depleted the excess PX capacity (See Figure 4). The global demand for PX has been growing at a quick pace and is expected to continue as the utilization of more PET bottle resins and polyester fibers find more uses in the global industry. Current projections from OrbiChem predict the average global consumption of PX to grow at 7% per year for the next 10 years [2]. Growth in PX consumption will be primarily driven by China's increased imports of PX, which is expected to be greater than 10 million tons per year [5].

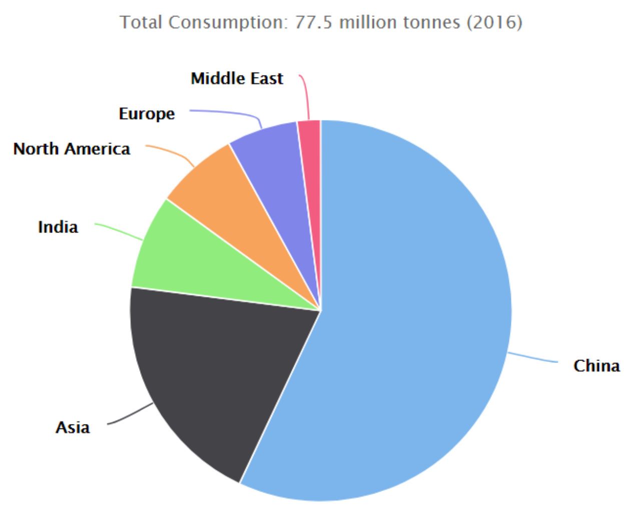


Figure 3: Demand of PTA per region. Market Demand in 2016: China (57%), Asia (20%), India (8%), North America (7%), Europe (6%), Middle East (2%). *Figure adapted from PlasticInsight.com* [2].

Feed Stock and Product Pricing

The production of paraxylene relies on a both naphtha and a methanol feed stock for the conversion of toluene to paraxylene. The valuable products and byproducts in the production of paraxylene include paraxylene, ortho-xylene, benzene, and hydrogen. The most important metric to focus on is the price spread between the naphtha feedstock, methanol price, and the selling price of the main product, paraxylene. Current prices show that naphtha sells for $500 USD per metric ton, methanol from Methanex corporation sells for $335 USD per metric ton, and paraxylene sells for $1050 USD per metric ton [5,6]. The cost spread between the naphtha feedstock/methanol feed stock and PX is upwards of $215 USD per metric ton, which makes the production of PX still quite profitable, even with the added expense of methanol. However, to get a more accurate price for the true feed of the specific methylation of toluene process, toluene was found to be $0.80 $/kg [50]. This price was used for the economic calculations in the feed moving forward, instead of the naphtha price. Moving on, current 2020 market prices for the byproducts of our process show that Benzene sells for $645.33 USD per metric ton, ortho xylene sells for $947.99 USD per metric ton, and hydrogen sells for $700 USD per metric ton [5]. Prices have been slowly falling over the past few years as an oversupply of PX still exists and oil prices have remained low. Factoring in the added profits from the sale of the byproducts, this process can be very profitable, even in an over supplied economy. The oversupply in PX is quickly diminishing though, so plant utilization will only continue to increase from here on. More PX plants will have to be built to meet future supply and current plants may have to increase the conversion of naphtha to PX to increase profitability/production capacity.

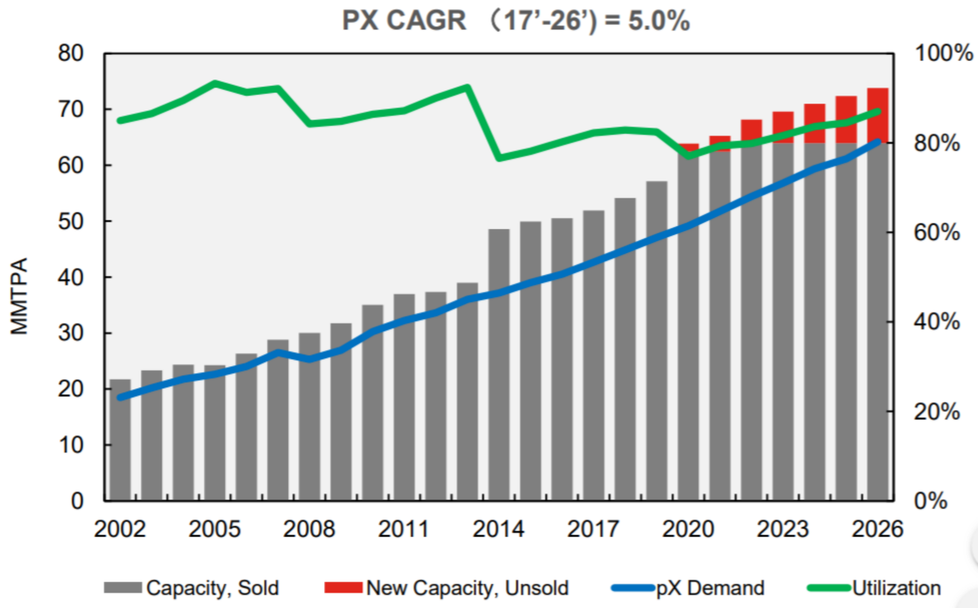


Figure 4: Global demand, capacity, and utilization of PX. *Figure adapted from Advances in Aromatics Technology by UOP Honeywell* [5].

Regulatory Environment

The production of paraxylene produces benzene as a valuable byproduct of the process. Benzene is an extremely carcinogenic chemical and in 2002, China imposed stricter regulations for permissible benzene concentrations in the air. They reduced the permissible level of benzene in the air from 10 ppm to 1.8 ppm [7]. Although these regulations are much more generous than in the United States, the stricter regulation will require higher costs associated with more safety equipment, safety inspections, and higher tolerances for equipment. To meet these regulations, higher capital costs can be accrued in the manufacture, storage, handling, and shipping of benzene. Other regulations imposed by China include capping the capacity on the amount of paraxylene produced in the region to maintain stability and profitably in the manufacturing of paraxylene. The paraxylene capacity cap fluctuates based on the current supply and demand. Currently most paraxylene plants can only operate at 60-80% capacity, depending on the output capacity of the plant [5]. We can expect that if we build a world class paraxylene production plant, we can expect to operate at 60% for the first few years and then slowly increase capacity as demand increases in China and oversupply begins to decrease.

Competitors

As of 2018, the world’s largest producers of PX were Hengli Petrochemical, Reliance Industries, SINOPEC, and ExxonMobil [8]. The world’s largest producer of PX is Hengli Petrochemical, which has a total output of 4.5 MMTPA. Reliance Industries is the second largest producer and currently has a facility that produces 4.3 MMTPA of PX. ExxonMobil currently has a facility that produces 3.5 MMTPA of PX. SINOPEC has a facility that produces 1 MMTPA of PX. It is predicted that by 2023, the Middle East will become the third largest producer of PX in the world. Their heavy investments into local feedstock integration, large-scale production facilitates, and access to low cost energy will make the Middle East very cost competitive on the global market [8]. The key to constructing a profitable PX process with the ability to compete with the world’s leading producers of PX will be to make the production of PX extremely cost competitive. This will include minimizing energy costs and maximizing the conversion of naphtha to paraxylene.

Goals and Objectives

After obtaining a more detailed look into the para-xylene industry from the market analysis, clear goals were drawn out to achieve a profitable world class para-xylene production plant. The purpose of this report is to provide an innovative and novel design for the production of para-xylene. The design will implement existing technological models, such as the methylation of toluene, while also incorporating new energy optimization ideas that will still allow for the production of para-xylene of optimal purity. Similar processes have already been modeled and commercialized by ExxonMobil, but by performing a heat integration of the highly exothermic reaction process, energy consumption will be reduced. Less energy consumption, along with lower capital costs from the avoidance of more complex separation equipment, will make the model more economically viable and more environmentally friendly. In addition to saving costs on lower utilities, less CO_2 will be emitted from the plant, helping the process fight back against pollution. Furthermore, this plant is designed to keep up with market demand and allow for the fluctuations of current product prices. For example, toluene and methanol have lower costs than what is made through the sale of para-xylene, making the methylation of toluene a good alternative to the more fundamental process of turning toluene into benzene.

Process Description

The conventional process of producing paraxylene involves the reforming of naphtha from petroleum distillate. Generally, the distillate is a narrow cut of naphtha, meaning most of the lighter hydrocarbons have been removed. The narrow cut of naphtha contains C_7^+ hydrocarbons which convert naphthene (a group of cyclic aliphatic hydrocarbons) to aromatic hydrocarbons through the reformation process [9]. Such processes are designed and operated to maximize the yield of the desired aromatic product, paraxylene, from the naphtha feed [9].

To begin the conventional process, naphtha in the reforming unit is contacted with a strong hydrogenation metal catalyst to produce reformate, hydrogen gas, and fuel gas. The reformate generally contains BTX, C_9^+ aromatic hydrocarbons, and ethyl benzene [9]. To obtain a purer liquid reformate, the hydrogen and fuel gas need to be removed, and are done so in the reforming and stabilizer units. The liquid reformate contains paraffins, ethyl benzene, BTX, and C_9^+ aromatic hydrocarbons, and is sent to a splitter to remove the C_9^+ aromatics. The product of the splitter, containing paraffins, ethyl benzene, and BTX, is then sent to an extraction unit in which a selective solvent is used to extract the undesirable aromatics and unconverted paraffins [9]. The product is a mixture of aromatic compounds composed of BTX, ethyl benzene, and small amounts C_9^+ aromatics. Next, the effluent mixture is sent to a fractionation unit in which benzene and toluene are removed as the distillate. The bottoms of the fractionation column contain xylenes and C_9^+ aromatics, which are then sent to a xylene column to separate out the desired para-xylene product [9]. The distillate is then fed to another column to separate benzene from toluene to obtain a benzene product. In bottoms of the benzene column, the toluene is sent to a trans alkylation / toluene disproportionation unit [9]. Benzene, xylenes and C_9^+ aromatics are further produced in the toluene disproportionation unit in the presence of hydrogen through the contact of an acid zeolite catalyst [9]. The effluent, containing mostly xylenes and C_9^+ aromatic hydrocarbons, are sent to a xylene rerun column where the heavy aromatics are removed, and the para-xylene rich aromatics are sent to a para-xylene recovery unit. In the para-xylene recovery unit, some para-xylene is recovered, while the para-xylene depleted stream is sent to an isomerization unit [9]. Fractional crystallization or selective adsorption (PAREX) is usually used to operate the para-xylene unit for separation. The para-xylene depleted stream is then fed to a xylene isomerization unit where the xylene isomers are isomerized back to their equilibrium concentrations. Furthermore, some of the aromatics are converted to ethane, benzene and diethyl benzene by isomerization. This is done to obtain a larger production of xylenes [9]. The equilibrium aromatic mixture is then recycled back to the para-xylene recovery unit after removal of any undesired C_9^+ aromatic by-products to further recover para-xylene [9].

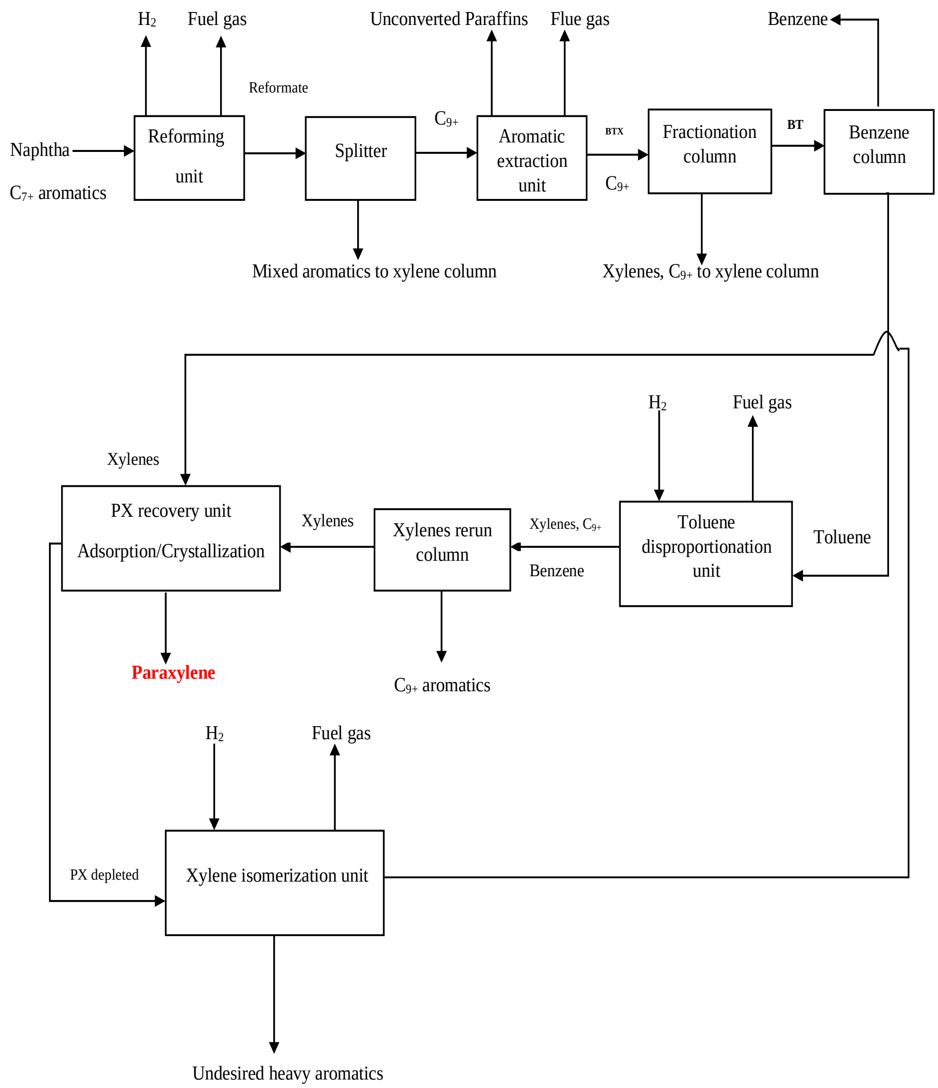


Figure 5: Block flow diagram for conventional paraxylene process from raffinate naphtha [10].

Possible solutions

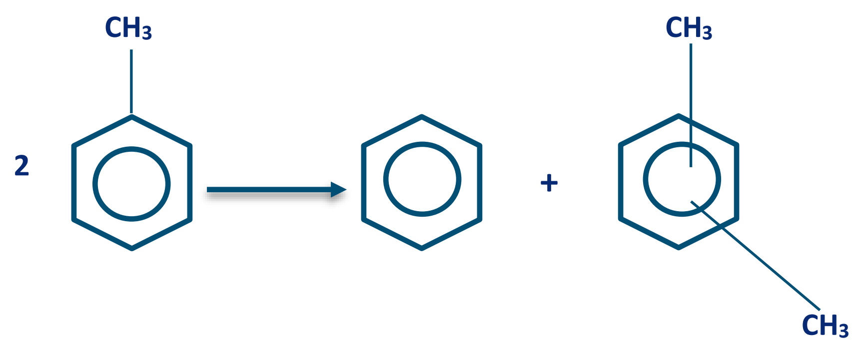
The production of para-xylene from naphtha feedstock requires a rigorous implementation of technological models to achieve a high-quality para-xylene product. A production method that produces high purities, high recoveries, and a flexible separation of para-xylene from its isomers is the most desirable operation. To achieve such a target, some of the competing industry methods are UOP Honeywell's process and ExxonMobil's process. Both processes operate in similar ways to the conventional process defined in the description above [see process description page 11]. However, differences show up in how toluene is converted into more valuable products and how para-xylene is separated out from its other isomers. UOP uses a toluene disproportionation process as a production method to convert toluene into benzene and xylenes. They also use a proprietary unit called PAREX as a means of separating the xylenes by a selective adsorption purification method. In contrast, ExxonMobil uses a toluene methylation production method involving alkylation of toluene with methanol in a bed reactor. The purification process is achieved through a reactive distillation unit. However, both UOP's and ExxonMobil's methods use a similar zeolite catalyst to convert toluene to para-xylene, making both processes more profitable [10].

1. UOP LLC

Production method:

In the case of toluene disproportionation, benzene and xylene are formed by a reaction of toluene over an acid zeolite catalyst [10]. The implementation of such technology requires three main processing areas. There is a reactor section where mixed aromatics and benzene are produced in the presence of hydrogen and a catalyst, a product distillation section where undesired aromatics are removed, and a para-xylene recovery section where xylenes are recovered and separated [10]. Crystallization or adsorption separation is typically used with this model to achieve at least 90% selectivity of paraxylene in the xylene mixture [10]. One of the disadvantages of these processes is that extensive separation is needed for high para-xylene purity due to low selectivity from the disproportionation unit. Moreover, such a process is only economically feasible when the demand of benzene is as high as that of xylene in the market [10].

The main toluene disproportionation reaction is the following:



Purification method:

UOP has developed and commercialized an adsorption purification method involving a proprietary solid selective adsorbent called ADS-37. This adsorbent is used in the PAREX process unit and is capable of achieving a purity of 99.9 wt% para-xylene at greater than 97% recovery [11]. Detailed information of the PAREX operation is not publicly available due to its sensitivity. However, separation of para-xylene from its isomers can be achieved by selective adsorption on a molecular sieve. This is possible due to the different positions of the methyl groups found in xylenes. The different positions of the methyl groups give different kinetic diameters: 6.7 A° for p-X, 7.3 A° for o-X and 7.4A° for m-X [10]. In other words, such differences result in a relatively high diffusion coefficient of para-xylene compared to those of other isomers[10]. Furthermore, separation is accomplished by exploiting the differences in affinity of the adsorbent for paraxylene relative to the other C_8 isomers on zeolites catalysts [10]. As a result, a recovery of 99.9 % pure para-xylene is possible in the extract while a mixture of its isomers is withdrawn in the raffinate [10].

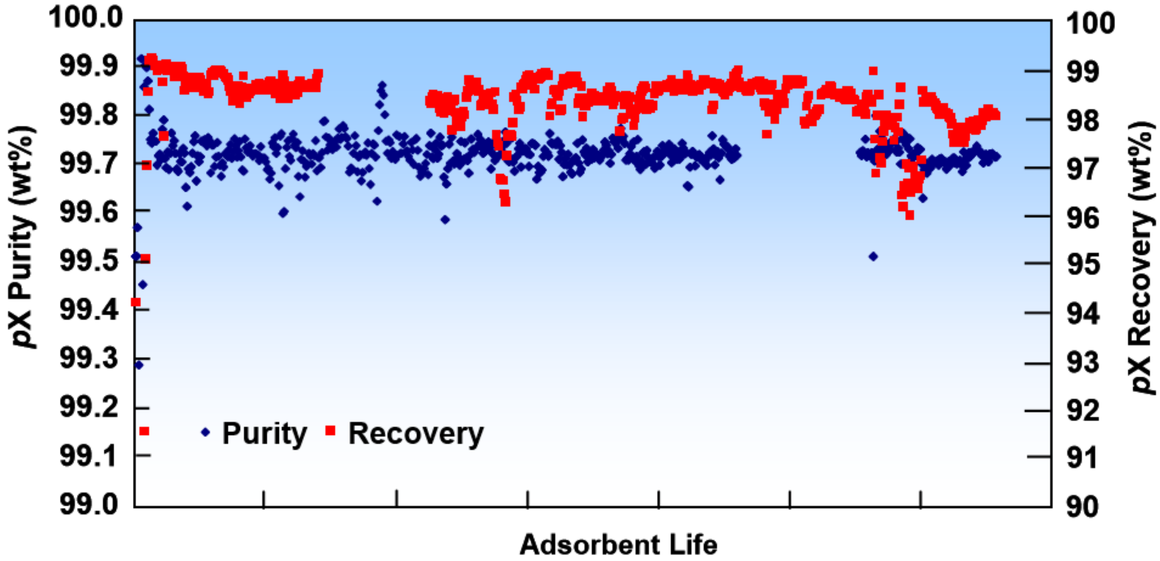


Figure 6: Commercial PAREX unit performance, in terms of purity and recover, with ADS-37 adsorbent for over 2+ years [11].

The PAREX unit facilitates the revamping of equipment to maximize production capacity. However, this makes the overall project long, with the cost of revamping being relatively high [11]. Hence, the UOP suggested process requires extensive implementation of complex units that would lead to high energy cost even though it results in highly pure para-xylene with high recovery.

Toluene methylation:

Production method:

Methylation of toluene is a more economically feasible process to produce para-xylene if the cost of methanol remains relatively low [10]. The main reaction used in this process consists of toluene and methanol over a zeolite catalyst ZSM-5. This reaction produces water and xylenes, with a thermodynamic equilibrium composition of xylene isomers being 23.55% p-xylene, 52.42% m-xylene and 24.03% o-xylene at around 400 ^{o}C [10]. However, different modifications of zeolite catalysts result in higher para-xylene selectivity which makes methylation a suitable alternative. Para-xylene selectivity of 90% or higher is possible with boron or magnesium doped zeolite catalyst [10]. In fact, boron modified catalyst exhibits a para-xylene selectivity higher than 99.9%, but the loss of boron from the catalyst during reaction makes it unstable [10]. Alternatively, a magnesium modified zeolite catalyst gives a para-xylene selectivity of 92.7%, while minimizing the contact time on external acid sites. This selectivity is achieved by the decrease in residence time, which prevents paraxylene isomerization [10]. It is possible to obtain an even lower residence time by adding N_2, or H_2O to the feed of the reactor, thereby increasing selectivity further.

Optimization of reactor operating conditions, such as pressure and temperature, can also increase para-xylene selectivity. The methanol is completely converted at a temperature ranging from 400 ^oC to 560 ^oC [10]. There are five reactions for the process of toluene methylation over Mg-modified ZSM-5 catalyst. This includes a main reaction, with four side reactions: toluene methylation, methanol dehydration, toluene disproportionation, paraxylene dealkylation and paraxylene isomerization. The rate expressions are given in terms of partial pressures ‘p’ and rate constants ‘k’.

1. **Toluene + Methanol $\longrightarrow$ p-Xylene + Water** $\quad r_1 = k_1 p_T p_M$
2. **2Methanol $\longrightarrow$ GH + 2Water** $\quad r_2 = k_2 p_M^2$
3. **Toluene $\longrightarrow$ $\frac{1}{2}$ Benzene + $\frac{1}{2}$ p-Xylene** $\quad r_3 = k_3 p_T$
4. **p-Xylene $\longrightarrow$ Toluene + $\frac{1}{2}$ GH** $\quad r_4 = k_4 p_{p\text{-}X}$
5. **p-Xylene $\longrightarrow$ $\frac{1}{2}$ m-Xylene+ $\frac{1}{2}$ o-Xylene** $\quad r_5 = k_5 p_{p\text{-}X}$

Where GH indicates gaseous hydrocarbons such as methane, ethane, ethylene, propane and butanes, which is represented as pure ethylene to ensure stoichiometric balance.

Toluene methylation is more favorable because its mild operating conditions, such as a pressure of 3-4 bar and a temperature of 400-500 °C, result in lower utilities [10]. Moreover, the more expensive implementation of separation units, such as the selective adsorption PAREX unit, are not needed for this method. Optimization of the process can be accomplished by analyzing the effects of parameters in such as feed composition, feed ratio, residence time, temperature and pressure. This work can be done in ASPEN plus using proper design specifications, property method and operating conditions, with the end goal of achieving a para-xylene selectivity as high as 99.7% [10]. All things being considered, the methylation of toluene production method is more advantageous in that it produces a higher yield that facilitates the separation of para-xylene from its isomers.

Purification method:

Conventional distillation processes for separating different components requires a significant difference in boiling points or a relative volatility of greater than 1. With this criteria, benzene and toluene can be traditionally separated due to relative volatility of 3.09. Para-xylene can't be separated though from its isomers using a simple distillation unit as their boiling points are very close and their relative volatilities are close to 1. Moreover, the presence of isomer components doesn't facilitate para-xylene separation due to the similar molecular structures of the isomers [10]. As a result, a rigorous distillation unit, such as reactive distillation column, is required to carry out a separation of paraxylene from m-xylene and o-xylene [10]. Alkylation of the xylene mixtures in a distillation unit makes separation of para-xylene from m-xylene possible [10]. Thus, m-xylene can selectively react with di-tertiary butyl-benzene (DTBB) and tertiary butyl-benzene (TBB) to form tertiary butyl m-xylene (TBMX) and benzene (B) [10]. Such reactions lead to a higher boiling point of tertiary butyl m-xylene that in turn becomes easier to separate from para-xylene through fractionation [10]. The normal boiling points of the individual components are shown below for comparison.

Table 1*:* Normal boiling points of components for the reactive distillation

Component	Normal boiling point, °C
p-xylene	**138.3**
m-xylene	**139.1**
o-xylene	**144.4**
TBB	**169.3**
TBMX	**205**
DTBB	**230**

The reaction in the reactive distillation column is rapid and immediately reaches equilibrium. The reactions are as follows:

$$\textbf{DTBB + m-X} \rightleftharpoons \textbf{TBMX + TBB}$$

$$\textbf{TBB + m-X} \rightleftharpoons \textbf{TBMX + B}$$

Benzene is quickly separated because it's the lightest of all components in the reactive distillation unit [10]. Sensitivity analysis can be used in an ASPEN simulation by implementing proper design specifications and varying parameters to achieve a paraxylene purity of 99.7%. Hence, this separation method is more favorable because it is conventional and avoids extensive energy consumption.

Optimal solution

For our design, we chose to implement the methylation of toluene method to produce para-xylene. In this method, an aromatic stream containing mostly toluene is preheated to a predetermined temperature, along with a fresh feed stream containing methanol. These streams are mixed, heated again, and then supplied to a reactor containing an alkylation catalyst. By controlling the feed stream temperatures, the molar ratio of methanol to toluene, and the temperature in the reactor, the selectivity of the reaction is maximized, along with the para-xylene product yield.

In the overall para-xylene production process, a liquid reformate is obtained from naphtha feed stock in a catalytic reforming unit. The reformate is then stabilized by expelling out the fuel gas in the mixture and sending the remaining aromatics to a splitter. The C_9^+ aromatics are removed in the bottoms of the splitter, while the tops are sent to an aromatics extraction unit. In the extraction unit unconverted aromatics of C_{20} and higher are removed at the top, along with some left over fuel gas impurities. The remaining BTX, ethyl benzene and C_{9+} impurities are sent to a benzene column. In the benzene column, benzene is removed as the distillate product, while the toluene rich stream is sent to a toluene methylation unit for further processing. In the toluene methylation unit, a toluene feed and a fresh methanol feed (which contains water to suppress side reactions) are first preheated and vaporized in a furnace before being sent to a fixed bed reactor for an alkylation reaction in the presence of magnesium doped ZSM-5 zeolite catalyst. In the alkylation reactor, toluene reacts with methanol to produce para-xylene. Side reactions also occur, such as methanol self-decomposition into gaseous hydrocarbons and water, and para-xylene conversion to m-xylene and o-xylene. The reactor effluent is then cooled down and sent to a flash drum where light compounds, such as gaseous hydrocarbons, are sent to the overhead. The bottom liquid is fed into a decanting vessel for further separation. In the decanting vessel, the upper layer product is water and the lower layer is an aromatic mixture containing benzene, toluene, p-xylene, m-xylene and o-xylene, among other heavier aromatics. The aromatic mixture then goes through a series of separations in which it is first sent to a benzene column, where the benzene distillate is recovered at the overhead. The bottoms, containing toluene and xylenes and aromatics, is then sent to a toluene column. Toluene is recovered as the distillate and recycled back to the reactor mixed feed. The bottom stream containing xylenes and mixed aromatics is then sent to a reactive distillation unit where 99.7 wt % of PX is recovered at the top of the reactive column [10]. In the reactive distillation columns, the xylene isomer m-xylene is reacted with ditertbutylbenzene in the presence of aluminum chloride to form tertbutyl xylene, which can be separated from p-xylene due to high volatility. It is sent to the top of the column while the remaining mixed aromatics is released at the bottom of the column [10].

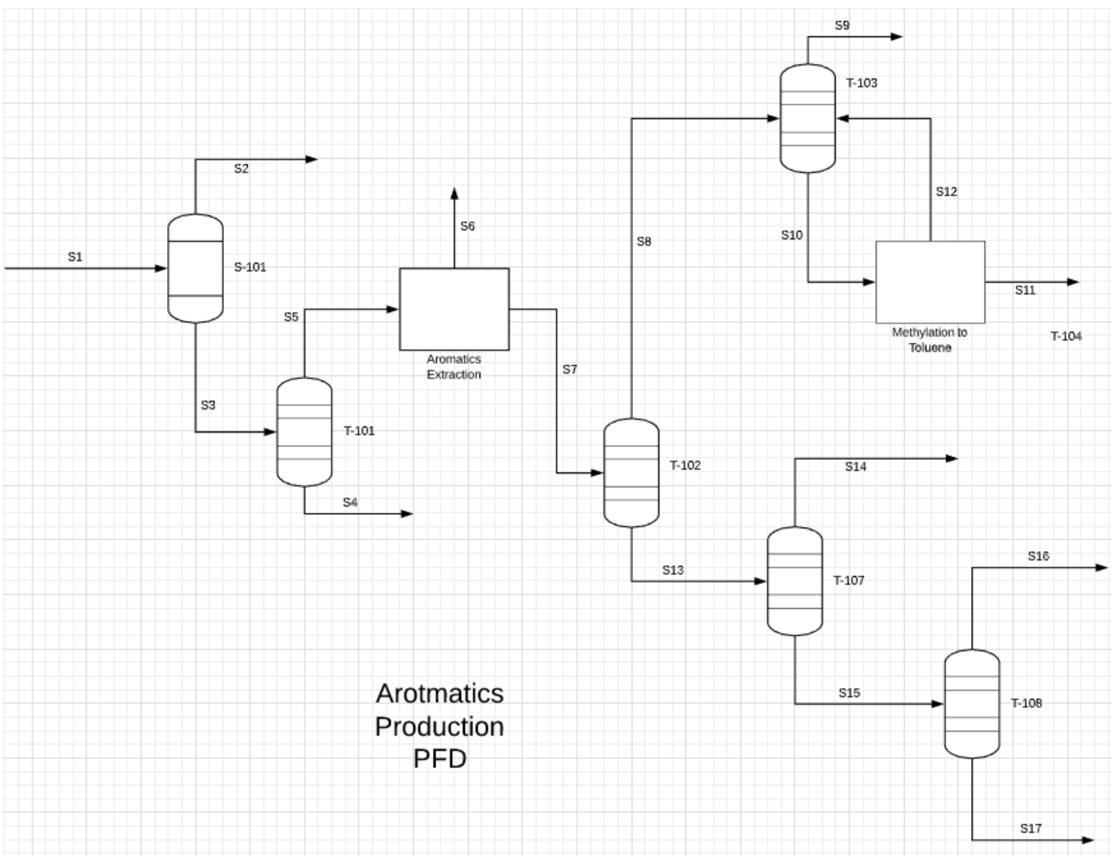


Figure 7: PFD of Aromatics Production through methylation of toluene process

The relative volatility of benzene / toluene and o-xylene / para-xylene were 3.09 and 1.17, respectively. Benzene can be easily separated from the mixture by distillation, and the separation of o-xylene requires a large amount of distillation due to its low relative volatility. It is difficult to separate para-xylene and meta-xylene by ordinary distillation, because the boiling points of the two components are too similar and their relative volatility is only 1.02. Due to the similar molecular structure, they are not affected by the presence of the third component, and extractive distillation does not work. Para-xylene and meta-xylene can be separated by reactive distillation, as reported by Saito et al. [12]. This method involves alkylating a mixture of xylenes. M-xylene reacts preferentially with di-t-butylbenzene (DTBB) and t-butylbenzene (TBB) to form t-butyl-m-xylene (TBMX) and benzene (B). The reaction is shown in Equations 1.1 and 1.2 [12]. M-xylene forms tertiary-butyl m-xylene with a higher boiling point, which can be easily separated from p-xylene by fractional distillation.

$$\text{DTBB} + \text{m-X} \longrightarrow \text{TBMX} + \text{TBB} \quad (1.1)$$

$$\text{TBB} + \text{m-X} \longrightarrow \text{TBMX} + \text{B} \quad (1.2)$$

The reaction occurs quickly and reaches equilibrium immediately [13,14]. The benzene produced in the alkylation reaction is the lightest and is distilled off, which can disrupt the equilibrium. Equilibrium data are given by Saito et al. [12] And the reactions in equations (1.1) and (1.2) are shown in equations (1.3) and (1.4), respectively.

$$K^R = x_{TBMX}\, x_{TBB} / \; x_{DTBB}\, x_{m\text{-}X} = 0.6 \quad (1.3)$$

$$K^R = x_{TBMX}\, x_B \,/\, x_{TBB}\, x_{m\text{-}X} = 0.16 \quad (1.4)$$

Toluene methylation is the most optimal para-xylene production process for it requires less complex and less extensive separation columns such as crystallization or adsorption (PAREX). The reason is that the methylation reaction exhibits a higher yield that facilitates the implementation of conventional distillation units. It is economically more viable and environmentally friendly than existing processes, considering that it uses less unit operations that contribute to lower utility costs.

Advantages for the Solution

High purity of para-xylene

One of the advantages of this technology is that it can produce xylene products rich in para-xylene through a bed reactor, thus greatly reducing the expectations and material consumption of the separation and alternation process. Alkylation of toluene and methanol is a new process for para-xylene production with greater economic benefits, which can maximize the utilization of toluene. The toluene alkylation method can maximize the production of valuable para-xylene products while reducing feed and energy costs. The high paraxylene selective process is based on commercially proven technology. Unlike other toluene conversion processes, it has no benzene by-products and no co-feed of hydrogen. It is the only method that can adjust the methyl ring ratio according to market conditions and allow unrestricted benzene co-feed to produce para-xylene [15].

Less equipment

In this method, the use of the distillation apparatus can reduce the use of the toluene disproportionation unit and parex unit. These two units are very expensive, so this a more economical apparatus. Furthermore, it has lower feed and energy costs [15]. In the distillation unit, the monitoring equipment of temperature and pressure is very sensitive. Therefore, we can detect and prevent hazards at quicker rates. The process is more stable and has consistent product yields.

Environment friendly

During the production of para-xylene process, no benzene co-product and hydrogen co-feed are produced. As a result, this is an environmentally friendly process. Because the solution has continuous regeneration of para-xylene, it has low sensitivity to poisons.

Process Safety and Environmental Hazards

Environmental and Safety Concerns

In the production of paraxylene through the methylation of toluene, the chemicals that are used in this process are outlined in the **Table 2** below.

Table 2: List of chemicals used and their physical properties in the production of paraxylene through the methylation of toluene. The majority of the chemicals used are flammable. [36-46]

Components	Boiling Point	Flash Point	Auto-Ignition	Flammability	OSHA PSM Highly Hazardous Material
p-Xylene	138.4 °C	25.0 °C	529.0 °C	UEL: 7% (V) LEL: 1.1% (V)	No
m-Xylene	139 °C	25.0 °C	528.0 °C	UEL: 7% (V) LEL: 1.1% (V)	No
o-Xylene	144 °C	31.0 °C	464.0 °C	UEL: 6.7% (V) LEL: 0.9% (V)	No
Benzene	80.1 °C	-11.0 °C	498.0 °C	UEL: 7.8% (V) LEL: 1.2% (V)	No
Toluene	110.6 °C	4.0 °C	535.0 °C	UEL: 7.1% (V) LEL: 1.2% (V)	No
Ethyl Benzene	136 °C	22.0 °C	430.0 °C	UEL: 6.7% (V) LEL: 1.0% (V)	No
Methanol	64.7 °C	9.7 °C	455.0 °C	UEL: 44% (V) LEL: 5.5% (V)	No
C9+ Aromatics (Trimethylbenzene)	169 °C	48.0 °C	515.0 °C	UEL: 6.4% (V) LEL: 0.9% (V)	No
DTBB	236 °C	84.0 °C	84.0 °C	N/A	No
TBB	169 °C	34.0 °C	450.0 °C	UEL: 5.7% (V) LEL: 0.8% (V)	No
TBMX	207 °C	68.0 °C	N/A	N/A	No

The following section will list the environmental issues associated with air emissions, water discharges for each chemical and possible mitigation strategies to help prevent the discharges. Process hazards will also be addressed for each chemical and safeguards will be discussed on how to address these hazards.

Note: The following sections outline the GHS category hazards. A GHS category of 1 is the most dangerous where a category of 4 is the least dangerous.

p-xylene, m-xylene, and o-xylene

Paraxylene is the main product of our process and is produced by the methylation of toluene in a packed bed reactor as well as being a component in the reformed naphtha feed. Paraxylene is flammable and toxic liquid at room temperature. The GHS has it assigned as a category 3 flammable liquid, a category 4 acute toxicity for inhalation and dermal absorption, a category 2 for skin irritation, and a category 2 for short-term (acute) aquatic hazards [36]. Paraxylene should be handled with care by the operators and the engineers during transport and production because it can irritate the eyes and skin if left untreated. If exposed to high levels of paraxylene vapors, it can lead to drowsiness and dizziness, potentially leading to workplace accidents/asphyxiation. The OSHA time weighted average (TWA) for paraxylene is 100 ppm. Higher chronic exposure to paraxylene has shown to have cariogenic effects.

Metaxylene is a by-product of our process and is produced by side reactions during the methylation of toluene in a packed bed reactor as well as being a component in the reformed naphtha feed. Metaxylene is flammable and toxic liquid at room temperature. The GHS has it assigned as a category 3 flammable liquid, a category 4 acute toxicity for dermal absorption, a category 2 for skin and eye irritation, a category 1 for an aspiration hazard for the respiratory system, a category 3 for short-term (acute) aquatic hazards, and a category 3 for long-term (chronic) aquatic hazards [37]. Metaxylene should be handled with care by the operators and the engineers during transport and production because it can irritate the eyes, skin, and respiratory system if left untreated. Breathing in a small amount of metaxylene vapors can cause drowsiness, dizziness, nausea, and headaches [37]. Breathing in a large amount of metaxylene vapors can lead to death. The OSHA time weighted average (TWA) for metaxylene is 100 ppm. Higher chronic exposure to metaxylene has shown to have cariogenic effects.

Orthoxylene is a by-product of our process and is produced by side reactions during the methylation of toluene in a packed bed reactor as well as being a component in the reformed naphtha feed. Orthoxylene is flammable and toxic liquid at room temperature. The GHS has it assigned as a category 3 flammable liquid, a category 4 acute toxicity for dermal absorption and inhalation, a category 2 for skin and eye irritation, a category 1 for an aspiration hazard for the respiratory system, a category 3 for short-term (acute) aquatic hazards, and a category 3 for long-term (chronic) aquatic hazards [38]. Orthoxylene should be handled with care by the operators and the engineers during transport and production because it can irritate the eyes, skin, and respiratory system if left untreated. Breathing in a small amount of orthoxylene vapors can cause drowsiness, dizziness, nausea, and headaches [38]. Breathing in a large amount of orthoxylene vapors can lead to death. The OSHA time weighted average (TWA) for orthoxylene is 100 ppm. Higher chronic exposure to orthoxylene has shown to have cariogenic effects.

Paraxylene has an autoignition temperature of 529 ^oC and in our process, the reactor effluent comes out at 525 ^oC. This presents a process hazard as we are very close to the autoignition temperature and if a runaway reaction occurred, this temperature could easily be reached. The composition of the vapor (67 v/v% of paraxylene) is much higher than the upper explosion limit (UEL) for paraxylene (7 v/v%) [36]. *Metaxylene* has an autoignition temperature of 528 ^oC and in our process, the reactor effluent comes out at 525 ^oC. This presents a process hazard as we are very close to the autoignition temperature and if a runaway reaction occurred, this temperature could easily be reached. The composition of the vapor (0.32 v/v% of orthoxylene) is much lower than the lower explosion limit (LEL) for metaxylene (1.1 v/v%) [37]. *Orthoxylene* has an autoignition temperature of 464 ^oC and in our process, the reactor effluent comes out at 525 ^oC. This presents a process hazard as we are way above the autoignition temperature. Since we operate the reactor at a pressure of 4 bar, this minimizes the risk for ignition as it raises the autoignition temperature. The composition of the vapor (0.2 v/v% of orthoxylene) is much lower than the LEL for orthoxylene (0.9 v/v%) [38]. The quantity of paraxylene stored on site will be 7,000,000 metric tons. The quantity of metaxylene stored on site will be 207,000 metric tons. The quantity of orthoxylene stored on site will be 126,000 metric tons.

Benzene, Toluene, Ethylbenzene

Benzene is a valuable by product of our process and is produced by side reactions during the methylation of toluene in a packed bed reactor as well as being a component in the reformed naphtha feed. Benzene is a flammable and highly toxic liquid at room temperature. The GHS has it assigned as a category 2 flammable liquid, a category 2 for skin and eye irritation, a category 1 for germ cell mutagenicity and carcinogenicity, a category 1 for blood toxicity and aspiration hazards, a category 2 for short-term (acute) aquatic hazards, and a category 3 for long-term (chronic) aquatic hazards [39]. Benzene should be handled with extreme care by the operators and the engineers during transport and production because it is highly flammable, toxic and is carcinogenic. If exposed to acute to low levels of benzene vapors, it can lead to drowsiness and dizziness; higher levels can lead to unconsciousness. Chronic exposure can lead to reproductive harm, blood cancer, and birth defects. The OSHA time weighted average (TWA) for benzene is 10 ppm.

Toluene is one of the reactants in the methylation of toluene reaction and reacts in a packed bed reactor with methanol to produce paraxylene and some by products. Toluene is a flammable and highly toxic liquid at room temperature. The GHS has it assigned as a category 2 flammable liquid, a category 2 for skin irritation, a category 2 specific target organ toxicity of the central nervous system, a category 1 for aspiration hazards, a category 2 for short-term (acute) aquatic hazards, and a category 3 for long-term (chronic) aquatic hazards [40]. Toluene should be handled with extreme care by the operators and the engineers during transport and production because it is highly flammable and toxic. If exposed to acute to low levels of toluene vapors, it can lead to drowsiness, dizziness, blurred vision, and muscle spasms; higher levels can lead to unconsciousness. Chronic exposure can lead to memory loss, loss of appetite, and loss of coordination. The OSHA time weighted average (TWA) for toluene is 10 ppm.

Ethylbenzene is a byproduct of our process and is a component in the reformed naphtha feed. Ethyl benzene is flammable liquid at room temperature. The GHS has it assigned as a category 2 flammable liquid and a category 4 for acute toxicity for inhalation [41]. Ethylbenzene should be handled with care by the operators and the engineers during transport and production because of its flammability. If exposed to acute levels of ethylbenzene vapors, it can lead to dizziness, chest pain, and eye irritation; higher levels can lead to unconsciousness. The OSHA time weighted average (TWA) for benzene is 100 ppm.

Benzene has an autoignition temperature of 498 ^oC and in our process, the reactor effluent comes out at 525 ^oC. This presents a process hazard as we are way above the autoignition temperature. Since we operate the reactor at a pressure of 4 bar, this minimizes the risk for ignition as it raises the autoignition temperature. The composition of the vapor (1.03 v/v% of benzene) is lower than the LEL for benzene (1.2 v/v%) [39]. Since it is close though, the concentration of benzene must be monitored and adjusted to ensure is stays below the LEL. *Toluene* has an autoignition temperature of 535 ^oC and in our process, the reactor effluent comes out at 525 ^{o}C. This presents a process hazard as we are very close to the autoignition temperature. The composition of the vapor (23 v/v% of toluene) is much higher than the UEL for toluene (7.1 v/v%) [40]. Since it is close to the autoignition temperature though, the concentration of toluene and the temperature of the reactor must be monitored and adjusted to ensure is stays at safe levels. *Ethylbenzene* has an autoignition temperature of 430 ^{o}C and in our process, the reactor effluent comes out at 525 ^{o}C. This presents a process hazard as we are way above the autoignition temperature. Since we operate the reactor at a pressure of 4 bar, this minimizes the risk for ignition as it raises the autoignition temperature. The composition of the vapor (0.04 v/v% of ethylbenzene) is lower than the LEL for ethylbenzene (1.0 v/v%) [41]. Since it is a flammable vapor, the concentration of ethylbenzene and temperature of the reactor must be monitored/adjusted to ensure is stays within safe limits. The quantity of benzene stored on site will be 750,000 metric tons. The quantity of toluene stored on site will be 3,800,000 metric tons. The quantity of ethylbenzene stored on site will be 30,000 metric tons.

Methanol

Methanol is one of the reactants in the methylation of toluene reaction and reacts in a packed bed reactor with toluene to produce paraxylene and some by products. Methanol is flammable and is a liquid at room temperature. The GHS has it assigned as a category 2 flammable liquid, a category 3 for acute oral, inhalation and dermal toxicity, and a category 1 for specific target organ toxicity of the eyes [42]. Methanol should be handled with care by the operators and the engineers during transport and production because it is highly flammable. If acute or chronic exposure to methanol vapors occurs, it can lead to dizziness, headaches, nausea, or blurred vision [42]. The OSHA time weighted average (TWA) for methanol is 200 ppm.

Methanol has an autoignition temperature of 455 ^{o}C and in our process, the reactor effluent comes out at 525 ^{o}C. This presents a process hazard as we are way above the autoignition temperature. Since we operate the reactor at a pressure of 4 bar, this minimizes the risk for ignition as it raises the autoignition temperature. The composition of the vapor (0.38 v/v% of methanol) is lower than the LEL for benzene (5.5 v/v%) [42]. Since it is close though, the concentration of methanol and the temperature of the reactor must be monitored to ensure is stays below the LEL. The quantity of methanol stored on site will be 15,000,000 metric tons.

C9+ Aromatic (Trimethyl Benzene)

Trimethyl Benzene, our representative of the $C9^{+}$ aromatics in the reformed naphtha, is a byproduct from the separation processes on the reformed naphtha. Trimethyl benzene is flammable and is a liquid at room temperature. The GHS has it assigned as a category 3 flammable liquid, a category 4 for acute toxicity for inhalation, a category 2 for skin and eye irritation, a category3 for specific target organ toxicity of the respiratory system, a category 2 for short-term (acute) aquatic hazard, and a category 2 for long-term (chronic) aquatic hazard [43]. Trimethyl benzene should be handled with care by the operators and the engineers during transport and production because it is flammable and is toxic. Exposure to trimethyl benzene vapors can lead to dizziness, light headedness, or unconsciousness [43]. The OSHA time weighted average (TWA) for trimethyl benzene is 25 ppm.

Trimethyl benzene has an autoignition temperature of 515 °C and in our process, the reactor effluent comes out at 525 °C. This presents a process hazard as we are above the autoignition temperature. Since we operate the reactor at a pressure of 4 bar, this minimizes the risk for ignition as it raises the autoignition temperature. The composition of the vapor (0.04 v/v% of trimethyl benzene) is lower than the LEL for trimethyl benzene (0.9 v/v%) [43]. Since it is close though, the concentration of trimethyl benzene and the temperature of the reactor must be monitored to ensure is stays below the LEL. The quantity of trimethyl benzene stored on site will be 44,000,000 metric tons.

DTBB, TBB, and TBMX

DTBB is a solvent used in the reactive distillation column to help separate out orthoxylene and metaxylene from paraxylene. DTBB is a liquid at room temperature. The GHS has it assigned as a category 1 for short-term (acute) and long-term (chronic) aquatic hazard [44]. DTBB should be handled with care by the operators and the engineers during transport and production because its toxicity. Exposure to DTBB vapors can lead to dizziness, light headedness, or unconsciousness [44]. The OSHA time weighted average (TWA) for DTBB is 100 ppm.

TBB is a solvent used in the reactive distillation column to help separate out orthoxylene and metaxylene from paraxylene. TBB is a liquid at room temperature. The GHS has it assigned as a category 3 flammable liquid, category 2 for eye irritation, a category 2 for short-term (acute) and long-term (chronic) aquatic hazard [45]. TBB should be handled with care by the operators and the engineers during transport and production because its toxicity and flammability. Exposure to TBB vapors can lead to dizziness, light headedness, or unconsciousness [45]. The OSHA time weighted average (TWA) for TBB is 100 ppm.

TBMX is a product from the reaction in the reactive distillation column that helps separate out orthoxylene and metaxylene from paraxylene. TBMX is a liquid at room temperature. The GHS has it assigned as a category 4 flammable liquid [46]. TBMX should be handled with care by the operators and the engineers during transport and production because its toxicity. Exposure to TBMX vapors can lead to dizziness, light headedness, or unconsciousness [46]. The OSHA time weighted average (TWA) for TBMX is 100 ppm.

DTBB is not considered a flammable liquid and thus possess minimal process hazards in the reactive distillation column. *TBB* has an autoignition temperature of 450 °C and in our process, the reactive distillation column operates at 162.9 °C. TBB is below the autoignition temperature and its boiling point. *TBMX* can produce flammable vapors, sources of ignition around the reactive distillation column need to be removed to prevent the possible ignition of the vapors. The quantity of DTBB stored on site will be 1,000,000 metric tons. The quantity of TBB stored on site will be 390,000 metric tons. The quantity of TBMX stored on site will be 600,000 metric tons.

Environmental Issues

Paraxylene has an acute severe toxicity to aquatic life. Metaxylene and orthoxyelene both have an acute toxicity and chronic toxicity to aquatic life. Benzene and toluene have an acute toxicity and chronic toxicity to aquatic life acute. Ethylbenzene has no listed serious effects to aquatic life. Methanol has no listed acute toxicity or chronic toxicity to aquatic life acute. Trimethyl benzene is listed as having acute toxicity and chronic toxicity to aquatic life acute. DTBB, TBB, and TBMX have no listed toxicities with aquatic life. The xylene, benzene, toluene, and trimethyl benzene concentration must be effectively monitored if any water with these chemicals is being discharged to the environment. The EPA has set a limit of 100 µg/L of total xylenes in a water discharge to the environment [35]. The EPA has set a limit of 5 µg/L of total benzene and a limit of 100 µg/L of ethyl benzene and toluene in a water discharge to the environment [35]. The EPA has set a limit of 150 µg/L of methanol in a water discharge to the environment though [35]. The EPA has set a limit of 10 µg/L of trimethyl benzene in a water discharge to the environment [35]. The EPA has set a limit of 100 µg/L of DTBB, TBB, and TBMX in a water discharge to the environment [35]. In order to meet the EPA limits for all the chemical limits in the water discharge, samples will be taken from the water discharge from our plant every hour and analyzed for the concentration of these chemicals. If the any of the chemical concentrations is at or above the EPA limit, an emergency hydrocarbon scrubber will be implemented to treat the water while the source of the leak is investigated.

Employee and Operational Safety Measures

All personal working on site must wear hard hats, oxygen sensors, and googles around any of the equipment. Around any noisy equipment (motors, pumps, compressors, etc.) all personal are required to wear ear plugs. When touching any equipment, all personal will be required to wear gloves as well incase of toxic/harmful chemical contaminates as well as protection from heat and electrical shock. When performing maintenance on any equipment, proper lock-out/tag-out producers will be followed, the ventilator air exchange rate will be increased in all enclosed areas, all personal will be required to wear chemical suites and respirators, and all ignition sources in the area must be eliminated/mitigated to prevent fires or explosions. All employees will be required to complete physicals twice a year and after any accidents. This is to ensure the health of all employees and detect any affects due to long term exposure to the on-site chemicals.

If a person has been exposed to a high acute exposure of any of the aforementioned chemicals, bring the person into fresh air as soon as possible and have them examined by medical professionals. To ensure the safety of all people working in the plant, adequate personal protective equipment will be used when handling all the chemicals. Extra PPE may need to be used when handling the more volatile and toxic chemicals. Also, ventilators will provide 2-3 exchanges of air per hour within the plant to minimize the buildup of toxic/explosive chemical vapors. An air monitoring unit will be employed to sample the air every hour to check for possible leaks and levels of the chemicals in the air. If the levels of any of the chemicals reach within 10% of the lower explosive limit or 10% of the TWA, an alarm will sound and the ventilators will increase the air exchange rate to 3-4 exchanges of air per hour. Paraxylene, metaxylene, orthoxylene, benzene, toluene, ethylbenzene, methanol, trimethyl benzene, and DTBB will be stored under a nitrogen headspace to reduce the risk of explosion/fires.

Table 3: List of operation safety concerns and the proposed mitigation of the hazards.

Process	Main Safety Concerns & Proposed Hazard Mitigation
Packed Bed Reactor	1. Temperature of the reactor goes too high and reaches the autoignition temperature of the flammable chemicals. 2. Flammable and toxic gases leak from pipes into the environment.
	1. Very accurate temperature controllers, emergency air cooling, and double sensors for accuracy. 2. Air monitoring, proper ventilation, and removal of ignition sources from the area of potential leaks.
Methanol Column	1. Flammable and toxic gases leak from pipes into the environment.
	1. Air monitoring, proper ventilation, and removal of ignition sources from the area of potential leaks.
Benzene Column	1. Flammable and toxic gases leak from pipes into the environment.
	1. Air monitoring, proper ventilation, and removal of ignition sources from the area of potential leaks.
Toluene Column	1. Flammable and toxic gases leak from pipes into the environment.
	1. Air monitoring, proper ventilation, and removal of ignition sources from the area of potential leaks.
Reactive Distillation Column	1. Flammable and toxic gases leak from pipes into the environment.
	1. Air monitoring, proper ventilation, and removal of ignition sources from the area of potential leaks.
Paraxylene Column	1. Flammable and toxic gases leak from pipes into the environment.
	1. Air monitoring, proper ventilation, and removal of ignition sources from the area of potential leaks.

HAZOP Analysis

Table 4: HAZOP Analysis of Methanol Distillation Column (T-201)

Vessel: **Methanol Distillation Column(T-201)**					
Intention: Separates a mixture of Methanol and Water at 1.2bar and 102 °C					
Guide Word	Deviation	Cause	Consequences	Safeguards	Action
Line: Column					
LESS	Temp	• Change in feed composition • Temperature controller failure • Less steam flow	• Improper product quality • Pressure change	• Install low temp alarm • Install temperature indicator	• Perform inspection
MORE	Temp	• Change in feed composition • Temperature controller failure • More steam flow	• Melting/damage to pipes • Autoignition of chemicals • Pressure change	• Install high temp alarm • Install temperature indicator	• Automatic shutdown
LESS	Pressure	• Crack/leak in the column • Reboiler underheating the liquid • Pressure indicator control failure	• Vessel implosion • Toxic release of gas • Change in composition	• Install low pressure alarm • Install pressure indicator	• Air monitoring for leaks
MORE	Pressure	• Vapor pressure increases • Blockage in the distillate pipe • Pressure indicator control failure	• Vessel explosion • Change in composition • Product loss	• Install high pressure alarm • Install pressure indicator	• Install pressure relief valve
LESS	Level	• Less feed flow rate • Level indicator control failure	• Change in composition • Damage to pumps if column dries out	• Install low level alarm • Install level indicator	• Emergency shut off
MORE	Level	• Higher feed flow rate • Level indicator control failure	• Change in composition • Increase in column pressure	• Install high level alarm • Install level indicator	• Emergency shut off

Table 5: HAZOP Analysis of Methanol Distillation Column (T-201) Cont.

Vessel: **Methanol Distillation Column(T-201)**					
Intention: Separates a mixture of Methanol and Water at 1.2bar and 102 °C					
Guide Word	Deviation	Cause	Consequences	Safeguards	Action
Line: Bottoms stream					
NO	Flow	• Pipe blockage • Pipe leakage • Valve closed	• Column dry out • No operation • Possible dangerous concentration	• None	• Schedule maintenance • Shut down
LESS	Flow	• Pipe partially plagued • Control valves failure	• Column level decreases • Column dry out • Poor product quality	• Install low level alarm	• Perform maintenance on equipment • Adjust flow
MORE	Flow	• Valve is fully open • Control valves failure	• Increases in level • Column flooding • Lower temperature • Poor separation	• high level alarm	• Perform Inspection • Adjust flow

Table 6: HAZOP Analysis of Condenser (E-205)

Vessel: **Condenser (E-205)**					
Intention: condenses vapor fluid from top					
Guide Word	Deviation	Cause	Consequences	Safeguards	Action
Line: Coolant stream					
NO	Coolant Flow	• Valve fully closed • Pipe burst	• Loss of pressure control in column • No condensing	• Install Alarms	• Automatic shut down
LESS	Flow	• Coolant Pipe blockage/leakage • Valve partially open	• Low pressure in column • Ineffective condensing	• Install low alarms • Increase coolant flow	• Investigate equipment
MORE	Flow	• Control system failure • Valve fully open	• High pressure in column • Condenser damage	• Install high alarms • Adjust flow	• Repair equipment

Table 7: HAZOP Analysis of Reflux drum (D-202)

Vessel: **Reflux drum (D-202)**					
Intention: Separates the reflux and distillate streams					
Guide Word	Deviation	Cause	Consequences	Safeguards	Action
Line: Distillate stream					
NO	Flow	• Control valve fail • Pipe burst	• Loss of pressure control in column	• Install Alarms	• Automatic shut down
LESS	Flow	• Valve partially open • Pipe blockage and/or leakage	• Low drum level • Reflux drum dry out • Low quality product	• Install low level alarms • Increase flow	• Schedule maintenance
MORE	Flow	• Valve close fails • Control system failure	• High drum level • Reflux drum overfill • Off specs. product	• Install high level alarms • Decrease flow	• Schedule maintenance
REVERSE	Flow	• Pipe blockage • Reflux/Distillate control valve failure	• Backflow of fluid • Reflux drum damage • Poor composition	• Install flow alarms • Check valves	• Schedule maintenance • Shut down

Table 8: HAZOP Analysis of Reboiler (E-206)

Vessel: **Reboiler (E-206)**					
Intention: Boils fluid mixture at the bottom of column					
Guide Word	Deviation	Cause	Consequences	Safeguards	Action
Line: Hot fluid stream (steam)					
NO	Steam Flow	• Pipe clogged or burst • Control valve failure	• No Bottoms • Loss of temperature control	• Install alarms and sensors	• Automatic shut down
LESS	Steam Flow	• Pipe partially clogged/leaked • Valve open fail • Less steam pressure	• Low temperature in column • Poor separation	• Install low alarms and sensors • More steam pressure	• Repair equipment
MORE	Steam Flow	• Valve fully open • More steam pressure	• High column temperature • Reboiler damage	• Install high alarms and sensors	• Shut down and repair

Process Control

Control Objectives

Safety:

- Implement proper control elements to ensure safe operation in order to protect the wellbeing of plant personnel and equipment
- Control release of toxic chemicals in the environment in order to avoid harm to nearby communities

Quality:

- Ensure that product quality meets market specifications by maintaining desired compositions, physical properties and performance properties (i.e. purity, density etc...).

Smooth Operation and Production rate:

- Key variables in process streams should be maintained close to their desirable values to prevent disturbances to downstream units [47].
- Achieve the desired product output by controlling parameters of production accordingly.

Process Cost Minimization:

- Operate at the lowest production cost while satisfying all other objectives

Operational Constrains:

- Higher reactor temperature can lead to catalyst deactivation and onset of undesirable side reactions.
- Uncontrolled fluid flow rates and level in distillation column can result in flooding [49].

Main Process Parameters

Heat Exchanger: Temperature

Reactor: Temperature, Pressure, WHSV, Composition and Production rate

Distillation Columns: Temperature, Pressure, Levels and Composition

Decanter and Drums: Level

Control Strategy

The control strategy adopted for the paraxylene process is based on the degrees of freedom available and the sensitivity of the critical parameters of operation. For instance, the methanol distillation process has five independent input variables (through V1, V2, V3, V4 and V5) and five dependent output variables (T, P, L1, L2 and A or F) that lead to zero degrees of freedom. The potential controlled variables are identified above as process parameters and manipulated variables are stream flowrates. The control strategy also ensures material and energy balances by manipulating and controlling column variables [49]. The control logics used for the paraxylene process are feedback, feedforward, cascade and ratio controls. The feedback control is used to respond to a disturbance in the system that already deviated the control variable from its target value and eliminate the offset. The feedforward control is implemented to anticipate disturbances by measuring the outflow of the process and keep the process variable at the desired set point [48]. The cascade control is used in the case where an output of one process variable is used to control a set point of another dependent variable [49]. Cascade control facilitates elimination of offset caused by disturbances by controlling the usually slower primary variable which in turn brings the faster secondary variable to setpoint. Ratio control is used when it is necessary to maintain a constant ratio feed variable in order to achieve a desired output that depends on relation of feed parameters [47].

Control Systems

1. MHX and R-201

Table 8: Control loops around MHX and R-201

Equipment	Tag	Controlled Variable	Control element	Manipulated Variable	Disturbances	Control Logic
Heat exchanger	MHX	MHX outlet Temperature to reactor	Valve	Cooling water flow rate on shell side	Variations in inlet temperature and flow rates	Feedback Control
Reactor	R-201	Composition Product Pressure WHSV	P-201 P-202 Valve	Toluene feed flow rate Methanol feed flow rate	Variations in operating conditions	Ratio Control Feedback control
		Reactor Temperature	Valve	Reactor affluent Stream	Reactor feed flowrates	Feedforward Control

2. D-201, T-201 and D-202

Table 9: Control loops around D-201, T-201 and D-202

Equipment	Tag	Controlled Variable	Control element	Manipulated Variable	Disturbances	Control Logic
Decanting Vessel	D-201	Level	Valve on S7	S7 flow rate	Drop in process fluid pressure	Feedback Control
Methanol Column	T-201	Temperature	Valve on reboiler steam stream	Reboiler steam flow rate	Drop in reboiler steam pressure	Feedforward Control
		Level	Valve on bottom stream	Flow rate of bottom stream	Feed flowrates	Feedback Control
		Pressure Composition	Valve on Condenser CW stream	Condenser Cooling water flow rate	Column pressure drop	Feedback Control
			Valve on reflux stream	Reflux flow rate	Feed composition upsets	Cascade Control
Methanol Column Drum	D-202	Level	Valve on Distillate Stream	Distillate Flow rate	Feed flowrates	Feedback Control

3. T-202 and D-203

Table 10: Control loops around T-202 and D-203

Equipment	Tag	Controlled Variable	Control element	Manipulated Variable	Disturbances	Control Logic
Benzene Column	T-202	Temperature	Valve on reboiler steam stream	Reboiler steam flow rate	Drop in reboiler steam pressure	Feedforward Control
		Level	Valve on bottom stream	Flow rate of bottom stream	Feed flowrates	Feedback Control
		Pressure	Valve on Condenser CW stream	Condenser Cooling water flow rate	Column pressure drop	Feedback Control
		Composition	Valve on reflux stream	Reflux flow rate	Feed composition upsets	Cascade Control
Benzene Column Drum	D-203	Level	Valve on Distillate Stream	Distillate Flow rate	Feed flowrates	Feedback Control

4. T-203 and D-204

Table 11: Control loops around T-203 and D-204

Equipment	Tag	Controlled Variable	Control element	Manipulated Variable	Disturbances	Control Logic
Toluene Column	T-203	Temperature	Valve on reboiler steam stream	Reboiler steam flow rate	Drop in reboiler steam pressure	Feedforward Control
		Level	Valve on bottom stream	Flow rate of bottom stream	Feed flowrates	Feedback Control
		Pressure	Valve on Condenser CW stream	Condenser Cooling water flow rate	Column pressure drop	Feedback Control
		Composition	Valve on reflux stream	Reflux flow rate	Feed composition upsets	Cascade Control
Toluene Column Drum	D-204	Level	Valve on Distillate Stream	Distillate Flow rate	Feed flowrates	Feedback Control

5. T-204 and D-205

Table 12: Control loops around T-204 and D-205

Equipment	Tag	Controlled Variable	Control element	Manipulated Variable	Disturbances	Control Logic
Reactive Column	T-204	Reaction product	Valve on TBB, DTBB Stream	TBB, DTBB stream flowrate	Feed composition upsets	Feedback Control
		Temperature	Valve on reboiler steam stream	Reboiler steam flow rate	Drop in reboiler steam pressure	Feedforward Control
		Level	Valve on bottom stream	Flow rate of bottom stream	Feed flowrates	Feedback Control
		Pressure	Valve on Condenser CW stream	Condenser Cooling water flow rate	Column pressure drop	Feedback Control
		Composition	Valve on reflux stream	Reflux flow rate	Feed composition upsets	Cascade Control
Reactive Column Drum	D-205	Level	Valve on Distillate Stream	Distillate Flow rate	Feed flowrates	Feedback Control

6. T-205 and D-206

Table 13: Control loops around T-205 and D-206

Equipment	Tag	Controlled Variable	Control element	Manipulated Variable	Disturbances	Control Logic
Paraxylene Column	T-205	Temperature	Valve on reboiler steam stream	Reboiler steam flow rate	Drop in reboiler steam pressure	Feedforward Control
		Level	Valve on bottom stream	Flow rate of bottom stream	Feed flowrates	Feedback Control
		Pressure	Valve on Condenser CW stream	Condenser Cooling water flow rate	Column pressure drop	Feedback Control
		Composition	Valve on reflux stream	Reflux flow rate	Feed composition upsets	Cascade Control
Paraxylene Column Drum	D-206	Level	Valve on Distillate Stream	Distillate Flow rate	Feed flowrates	Feedback Control

Methanol Distillation Column(T-201):

The controls of the methanol column's operating conditions such as flow, temperature, pressure, level, and composition are implemented to ensure a safe operation and an achievement of desired recoveries of mostly methanol at the top and water at the bottom. The pressure of the distillation column is the most crucial variable in obtaining a safe and effective separation. The feed flow rate of the column is set by the level controller on the previous unit D-201(decanting vessel) and assumed to be constant.

The pressure of the column is controlled by manipulating the cooling water flowrate of the condenser E-205 through a valve V1 to avoid an under or over pressurized system and minimize energy consumption in the condenser. The lower limit of the column pressure is set at 1.5 bar and the upper limit at 2 bars. Such pressure limits are used so that water can be the coolant at ambient conditions. Moreover, separation of methanol from water can be easily achieved at such low pressures due to lack of azeotropes. High pressure alarm is put between the controller and transmitter for a quick response to avoid exceeding safety limitation. In the case of disturbance or over pressure, the transmitter sends a signal and alarms the controller which in turn brings the pressure to the desired setpoint of 1 bar by either automation or controller room operator.

The level of the reflux drum D-202 is kept in check by manipulating the flow rate of the distillate methanol. The lower limit of the drum level is set to be 20% by volume and the upper limit to be 80% by volume. A low and high alarms are set to notify the controller in case the level deviates from the setpoint of about 70% to avoid either drying or accumulation of fluid. Such actions prevent increase of utility cost in the condenser and ensures proper reflux and distillate outputs. The composition of the distillate product is maintained at the least purity of recyclable methanol by manipulating the reflux flow rate that goes back into the column through valve V3. The composition of the methanol is cascaded through a flow control that first receives a signal from the flow transmitter on the reflux stream. The output of such flow rate is then used to bring the composition to the desired setpoint through a flow indicating controller. The flow control also contributes to the change in temperature of the top part of the column. A low and high alarms are used to alert the controller to bring variables back to setpoint in case of disturbances. The lower limit of the methanol composition is set to be 98% as it is needed to be recycled back to the reactor feed and the set point is at 99%.

The temperature of the column is controlled by manipulating the steam flow rate of the reboiler E-206 through valve V4 to ensure that desired temperature is maintained for a safe and efficient separation. The temperature sensor is put at the column location where there is a significant temperature change so that the impacts are favorable for a good separation. The lower limit of the column temperature is set at 90 °C and the upper limit at 110 °C. Low and high temperature alarms are implemented so that the controller is alerted for action in case of setpoint deviation. The temperature sensor senses the disturbance and sends a signal to the controller through the transmitter in order to reestablish the setpoint of 102 °C. Such temperatures are chosen in accordance to the boiling points of the most prevalent components involved in the separation process.

The level of the column is controlled by manipulating the flow rate of the bottom product through valve V5 to maintain a proper column level for a safe and optimal separation. The lower limit of the column level is set at 30% by volume and the upper limit is at 90% by volume. Such limits as chosen to avoid the methanol column from drying at low level and overflooding at higher level. Low and high alarms are installed in the control room so that the process variable can be brought back to setpoint in case of disturbances. The level transmitter senses the deviations and signals the indicating controller which in turn regulates the level by either automation or manual operation from an operator.

Table 14: T-201 and D-202 Control Variable

Tag	Controlled Variable	Set Point	Lower Limit	Upper Limit	Manipulated Variable
T-201	Level	80%	30%	90%	Bottoms flowrate
	Temperature	102 °C	90 °C	110 °C	Steam flowrate
	Pressure	1.5 bars	1 bar	2 bars	Coolant flowrate
	Composition	99%	98%	100%	Reflux flowrate
D-202	Level	70%	20%	80%	Distillate flowrate

Process Control Loop schematics:

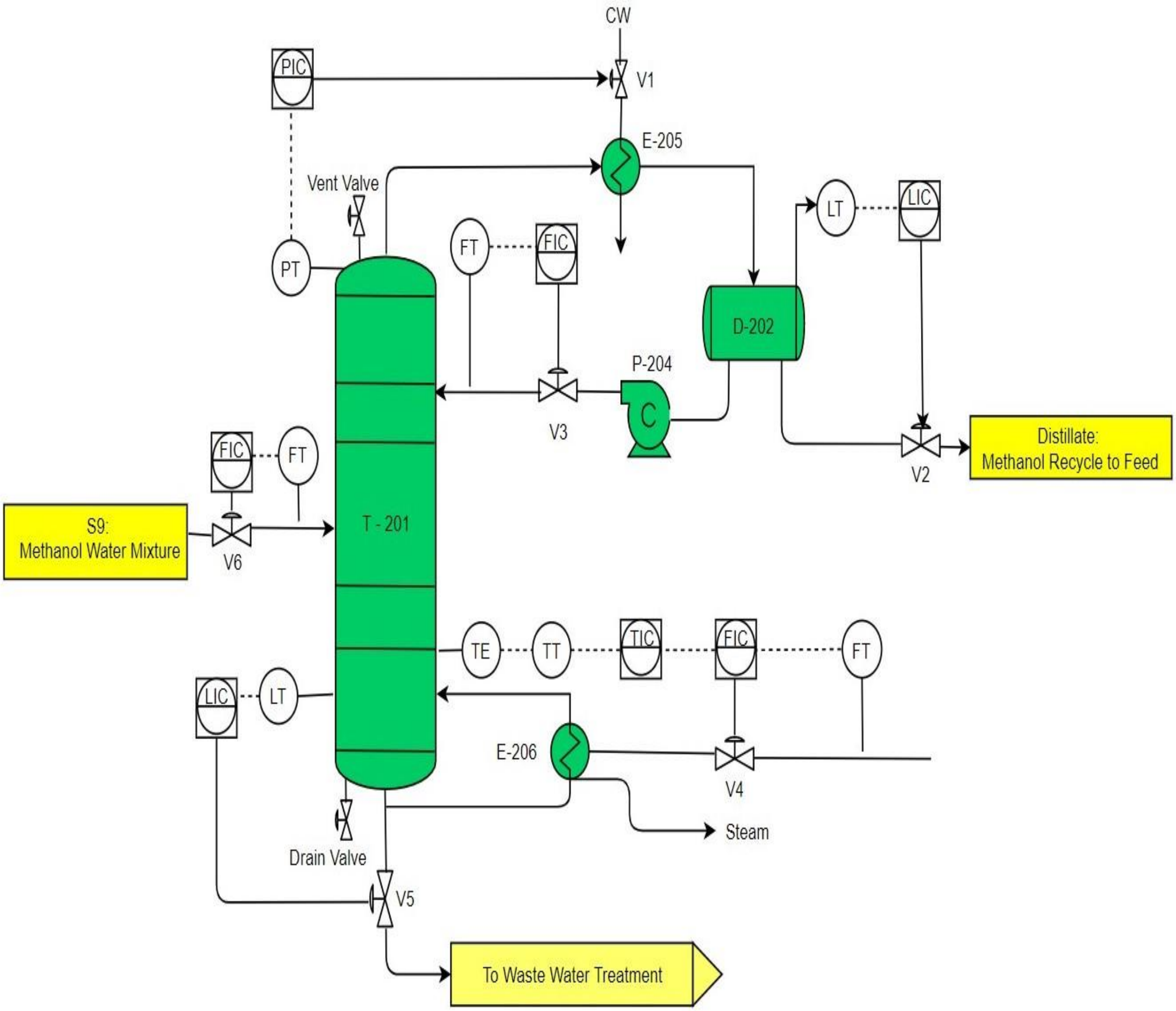


Figure 8: Methanol Distillation column (T-201) control scheme Before HAZOP

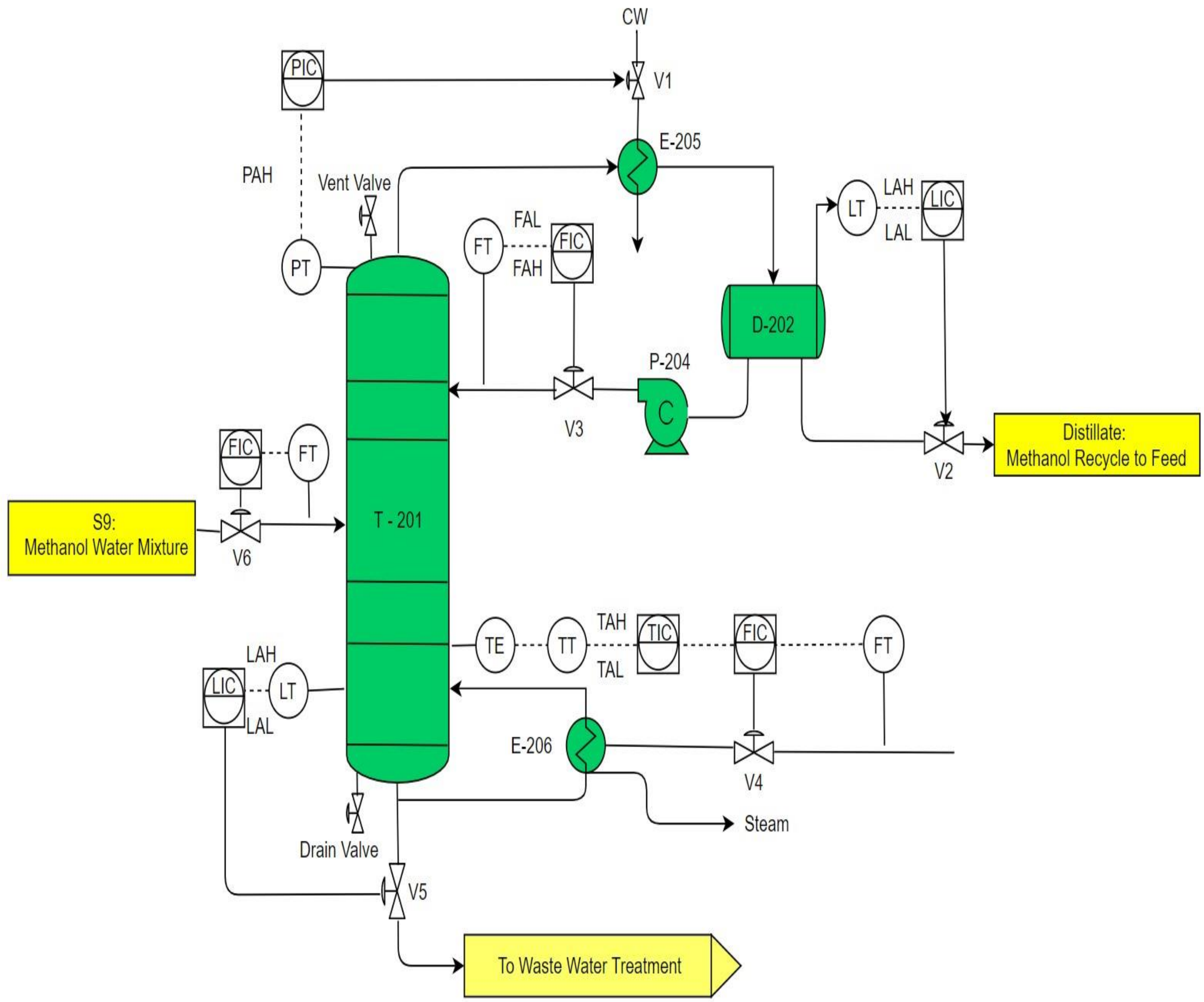


Figure 9: Methanol Distillation column (T-201) control scheme After HAZOP

Economic Analysis

ISBL

The economic analysis for the methylation of toluene process was based on the fundamentals and guidelines set in Gavin Towler's book: *Chemical Engineering Design*. This set the foundation on how to calculate important values such the inside battery limits costs, outside battery limits cost, variable costs of production, fixed costs of production, and much more. Excel was the chosen software to be used in the calculation of these costs, and its pages details can be found in appendix A8.

The first economic parameter calculated was the ISBL, or inside battery limits investment. These investments include all of the process equipment needed for a new plant, as well as the installation fees. In addition to all the process equipment, bulk items were also included, such as the catalysts for the project. Civil works costs and other miscellaneous fees could have been added to this number, but it was decided to be out of the scope for this report. The final ISBL calculated was $38.28 MM.

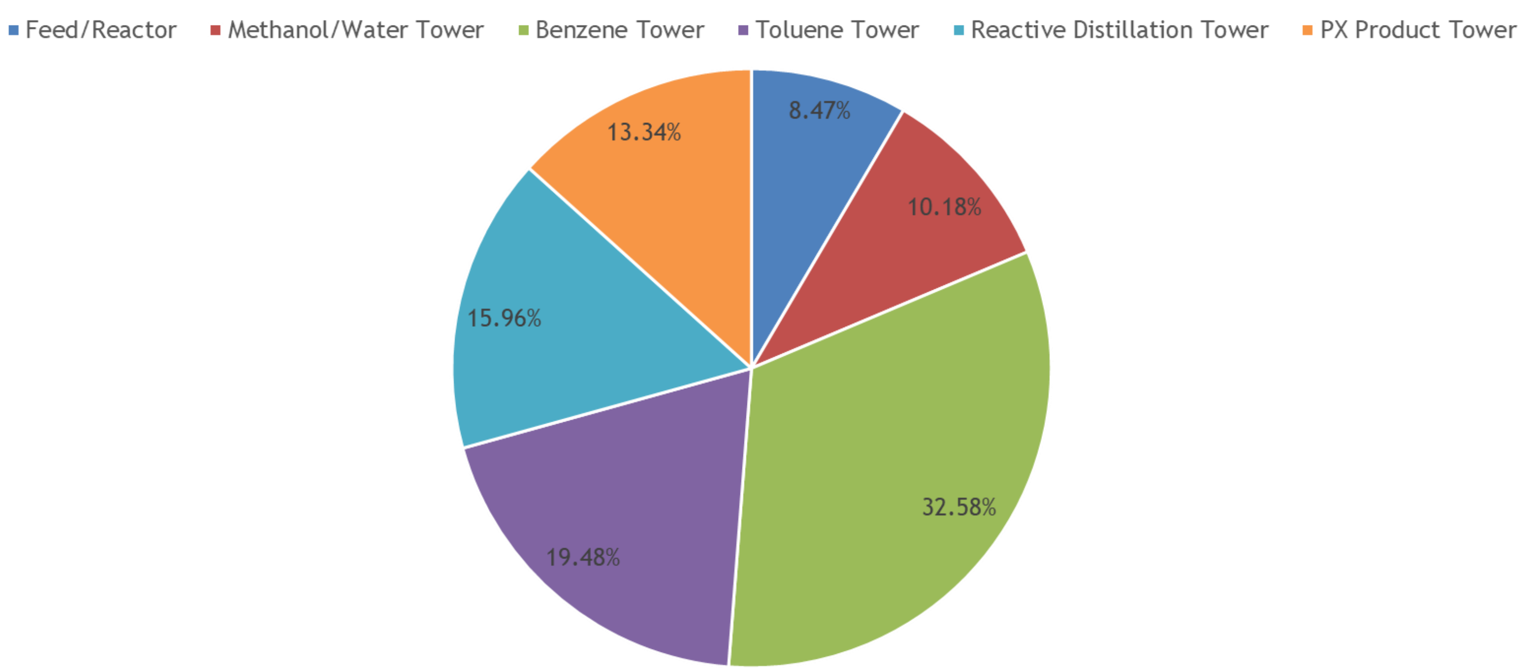


Figure 10: ISBL distribution for the different sections in the methylation of toluene.

The ISBL distribution is weighted pretty similarly for every portion of the process except the benzene section. The benzene section accounts for almost 1/3 of the ISBL costs. This high cost is due to the benzene distillation column being the biggest in the process at 50 sieve trays and a diameter of 18.5 meters. For future economic analysis down the road, it would be beneficial to do a sensitivity analysis for this column. The goal would be to try and reduce the numbers of trays, along with reducing the size of its diameter, while still obtaining the desired compositions of the distillate and bottoms products. This could potentially have a great impact on the total ISBL costs of the project.

Process Equipment Calculation and Comparison

The ISBL included pieces of process equipment whose specific costs needed to be calculated first. There were two main methods available to accomplish these cost calculations: through the equations given by Towler [27] or through the *ASPEN Process Economic Analyzer*. For a comparison between methods, both were used to calculate the cost of the methanol tower (T-201). For ease of use, the location for this distillation column was set in the gulf coast (default setting for Towler). However, for the chosen method explained later, the location was set for indigenous China which lowered overall costs for the plant.

For the method outlined by Towler, the following equation was used:

$$C_e = a + bS^n$$

Where C_e is the equipment cost on the gulf coast in January 2006, a and b are cost constants, S is the size parameter, and n is the exponent for that type of equipment.

The size parameter used was taken from table A5-1A in appendix 5 and the other parameters were taken from Table 6.6 in Towler's book [27]. For T-201 the purchased cost came out to be $1.1 MM and the installed cost at $4.41 MM. Using the same column dimensions and sizing in ASPEN, the purchased cost (*figure 11*) came out to be $1.72 MM and the installed cost at $3.89 MM. Both of these costs were very similar, giving confidence to the hand calculations.

Summary Costs

Item	Material(USD)	Manpower(USD)	Manhours
Equipment&Setting	1716800.	59183.	1910
Piping	865516.	551290.	17962
Civil	70958.	56598.	2283
Structural Steel	61310.	10054.	356
Instrumentation	143633.	23578.	765
Electrical	3590.	1906.	65
Insulation	160822.	148125.	6496
Paint	5674.	11787.	521
Subtotal	**3028303**	**862521**	**30358**

Total material and manpower cost=USD 3890800.

Figure 11: Summary of T-201 costs calculated in ASPEN Process Economic Analyzer

With both calculations being similar, the hand calculation method was chosen to be used to complete the equipment costs. This method was chosen based on the costs being slightly higher than the ASPEN costs. For this process, it was chosen for the design to account for higher costs, and potentially save money with cheaper equipment down the road, than have to end up paying more for the equipment later. Also, since Towler's methods were used in all of the other economic indicators, Towler's method was chosen for continuity purposes.

OSBL, Engineering Costs, Contingency, and Working Capital

The following costs were calculated as a percentage of the ISBL: outside battery limits investment, engineering costs, contingency, and working capital. The OSBL, or offsite costs, include the add-ons that must be made to the process plant's infrastructure to accommodate a new plant. This could include electric main substations, water pipes, and maintenance facilities. These costs were calculated at 40% of the ISBL and came out to be $15.31 MM. The engineering costs accounted for the engineering design costs needed to carry out the project. This value came out to be $7.66 MM, 20% of the ISBL. The contingency costs included any extra costs that were projected to come up later in the project and were calculated to be $11.48 MM at 30% of the ISBL. Lastly, the working capital was calculated to be $5.74 MM at 15% of the ISBL. This value accounted for the costs needed to start up the plant, and to continue to run it until profits are made. The total fixed capital cost is the addition of the ISBL, OSBL, engineering costs, contingency, and working capital, and was calculated to be $78.48 MM.

Variable Costs of Production

The variable costs calculated for this project correspond to the costs proportional to the output capacity of the plant. For the scope of this report, this cost included the price of raw materials, utilities, and consumables (solvents). However, Towler used the profits from the by-products to offset these costs [27]. Therefore, the profits from the benzene ($0.65/kg) and ortho-xylene ($0.95/kg) by-products are seen here. The following equation was used to calculate the VCOP:

$$VCOP = Raw\ Materials + Utilities + Consumables - ByProducts$$

The raw materials used in our process were a toluene feedstock ($0.80/kg) and methanol ($0.34/kg) feedstock. The utilities used were low pressure and high pressure steam, electricity, and cooling water. The consumables used were DTBB ($20/kg) and TBB ($20/kg), the solvents needed for the reactive distillation column T-204 [32]. The VCOP was calculated to be $3,065.39 MM/year, not including the utilities. With utilities the VCOP was calculated to be $3,144.93 MM/year.

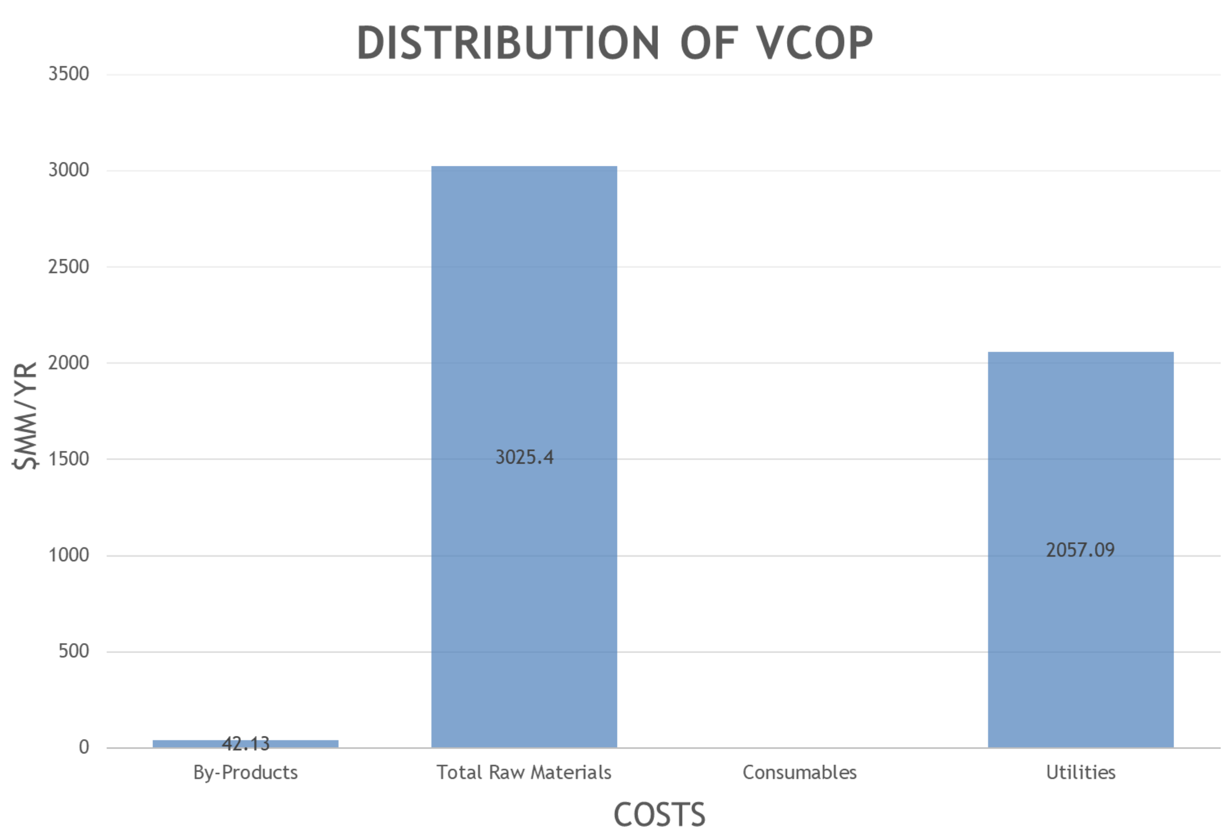


Figure 12: Distribution of the variable costs of production in millions.

The VCOP is unevenly distributed in which the raw material costs and utilities costs both make up about most of the total. With the desired output, the raw materials needed for the process cannot change. Therefore, the only analysis that can be done to reduce the total VCOP would be in the utilities sector. This gives precedence to the heat integration that will be done in the final report for this project. If utilities can be optimized, millions of dollars can be saved for this process.

Fixed Costs of Production

The fixed costs of production include worker compensation, maintenance, and overhead expenses. The worker compensation is estimated at 4 total shifts, 4.8 operators per shift, and $30,000/year salaries. This salary was chosen based on the average chemical engineer salary in China [33]. Supervision is also needed for the plant and was calculated to be 25% of the labor costs. Direct overhead was also calculated at 45% of the labor costs.

Table 15: Worker compensation for the plant.

Costs	$M/yr
Labor	576.00
Supervision (25% of labor)	144.00
Direct Overhead (45% of labor)	324.00

Maintenance and overhead expenses were also calculated as percentages of ISBL (corresponding to maintenance of the equipment priced out in the ISBL), labor, and fixed capital costs.

Table 16: Maintenance and overhead expenses for the plant.

Costs	$M/yr
Maintenance (3% of ISBL)	2,354.26
Plant Overhead (65% of labor and maintenance)	2,208.87
Tax and Insurance (2% of fixed capital)	1,569.51

The addition of all of these costs in tables 15 and 16 correspond to the total fixed cost of production at $7.18 MM/year.

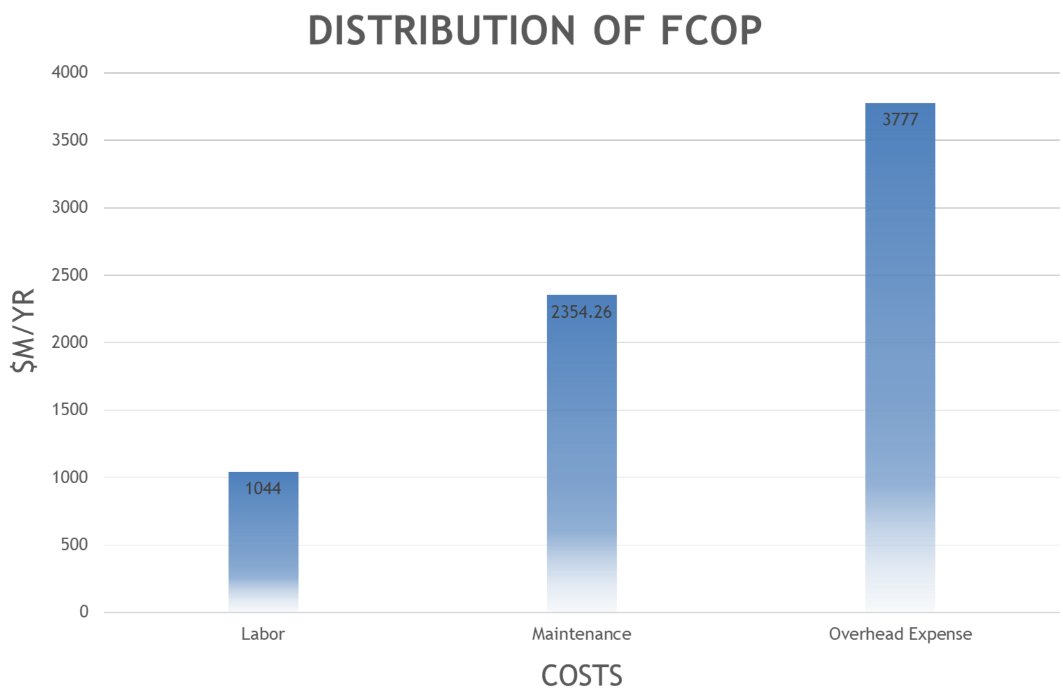


Figure 13: Distribution of the FCOP for the methylation of toluene plant.

The majority of the FCOP comes from the overhead expenses of the plant. This number is so large because it includes a percentage of labor, maintenance, and total fixed capital. The maintenance and total fixed capital percentages relate directly back to the ISBL, and therefore can be lowered if the cost of the equipment is lowered.

Net Present Value, Simple Payback Period, and ROI

The net present value of a project is used as an economic indicator for its future success and viability. It is a function of interest rate and the time period in which it is studied. NPV is often used because it accounts for the time value of money, as well as annual variations in expenses and revenues [27]. The NPV for this project was calculated with a 2-year construction period, a 20-year plant life, a 7 MACRS recovery term, a tax rate of 25% (based on the average tax rate for corporations in China [34]) and with an interest rate of 8%.

At 10 years, the NPV was calculated to be $144.8 MM/yr. At 20 years, the NPV was calculated to be $219.10 MM/yr. These numbers are so large because of the yearly revenue acquired each year, explained in the next section. The DCFROR was calculated to be 92% and is the highest possible interest rate that will allow the project to break even at year 20. This number is also very high and is also related to the high revenue generated each year.

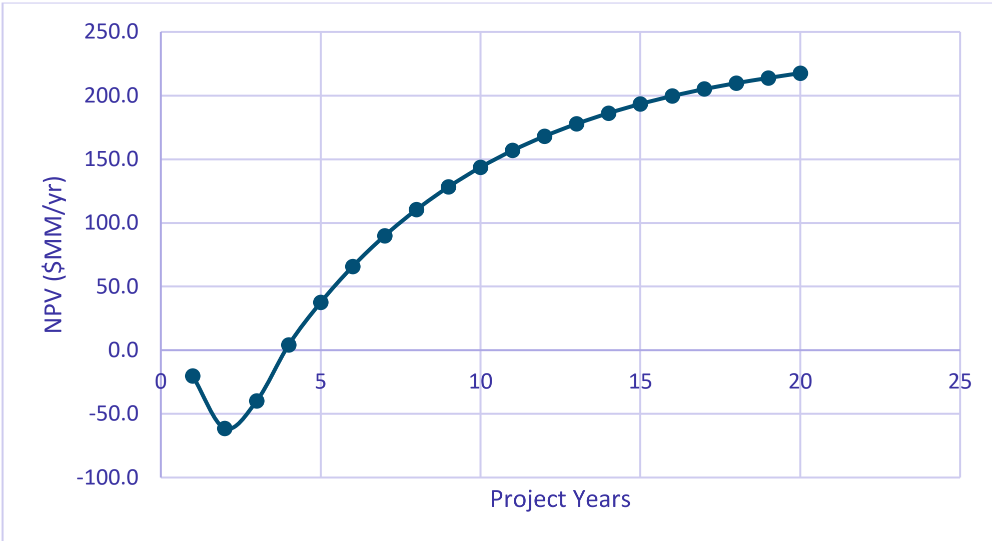


Figure 14: NPV for the lifespan of the project.

The NPV is used as a better economic indicator than ROI and the simple payback period as explained in the paragraph above. However, even though these values don't account for the time value of money, they were also calculated. The ROI of the project is calculated to be 66% at 10 years and 78% at 15 years. The simple payback period was calculated to be 1.27 years.

Revenues and Gross Profits

The revenues generated from our process are gained from the incomes earned from the sale of the products. The main product sold from our process is para-xylene, which is sold at a price of $1050 USD per metric ton [5,6]. With the side product incomes accounted for in the variable costs of production, para-xylene is the only product considered in the total revenue. For the given capacity of the process stated in the design basis in appendix 1, the total revenue is calculated to be $3,072.56 MM/yr.

To solve for the gross profits, the following equation is used:

$$Gross\ Profit = Revenue - \ Cash\ Cost\ of\ Production$$

Where the cash cost of production is the addition of VCOP and FCOP

The gross profit is calculated to be $85.60 MM/yr.

Sensitivity Analysis

For any given process, the economic analysis (including the analysis in this report) is only an estimate based on the total investment needed for the project and the total revenues estimated for the projects stated capacity. Therefore, any changes in pricing can have a potentially dramatic effect on the project's economics. To account for possible fluctuations in the project's lifespan, a sensitivity analysis can be done to see how it could be affected down the road. The net present value is used as the economic indicator for this analysis, while the ISBL, OSBL, raw material cost, product price, total fixed cost, and utility parameters are fluctuated.

The first sensitivity analysis performed was done by changing the product prices and raw material costs.

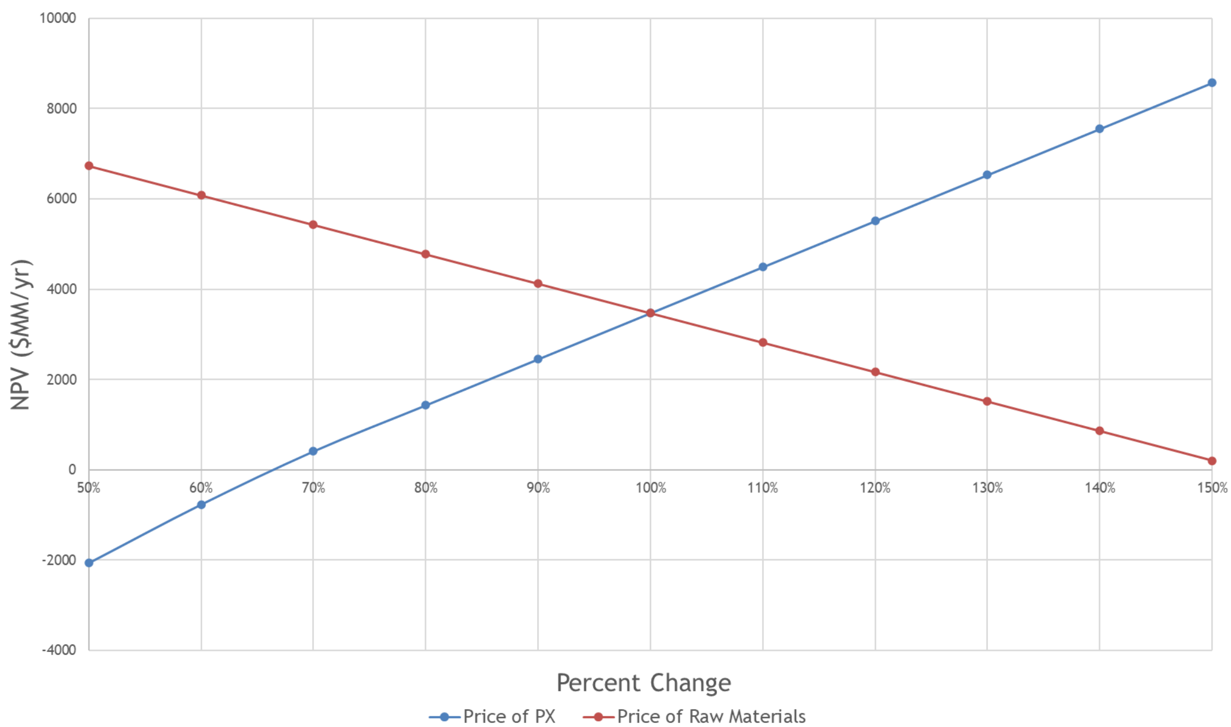


Figure 15: Changes in product and raw material costs, seen in the final NPV at year 20.

The changes in raw material and product pricing had the greatest effects on the NPV of the project. The change in the products price had the biggest range of NPV change from -$2,067 MM/yr to $8,567 MM/yr. It also had the steepest slope, meaning it was the most sensitive to fluctuations in pricing. This is due to the enormous output of 4.5 MMTPA for the plant. With such a large output accounting for such a large revenue, the price fluctuation can dramatically affect the final NPV. Though not as sensitive, the price of the raw materials needed for the process can also greatly affect the final NPV.

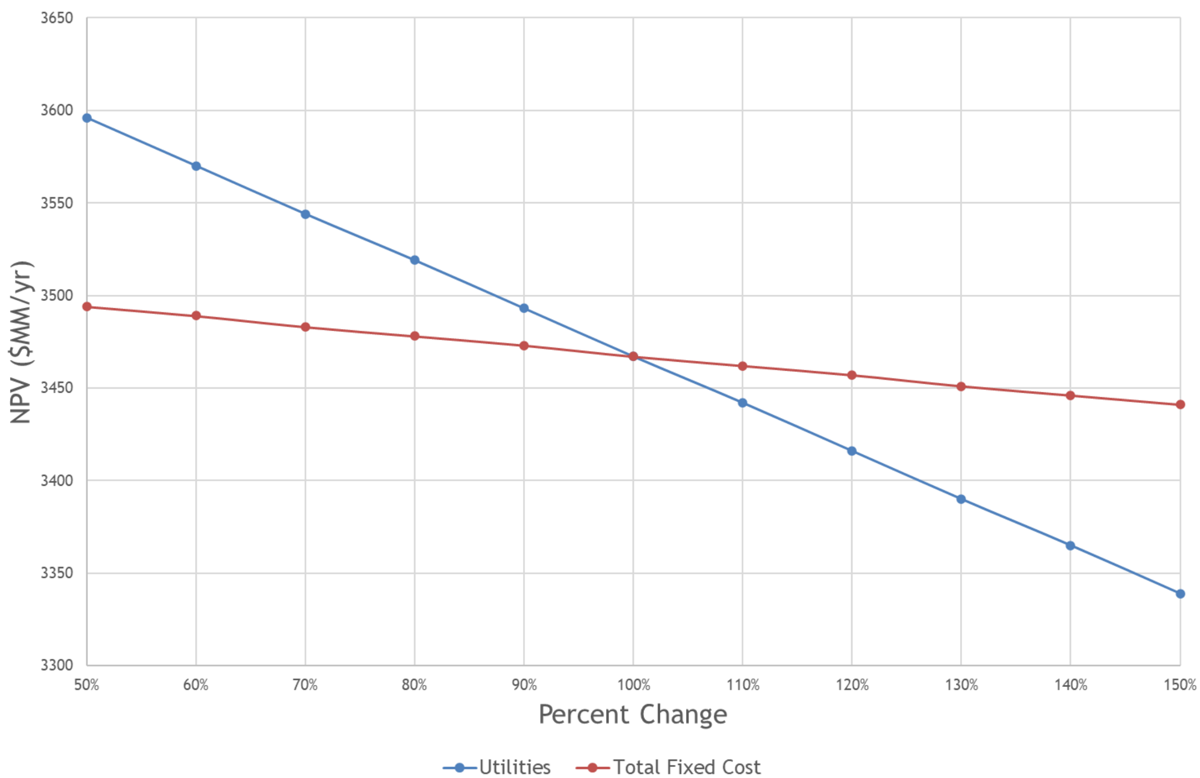


Figure 16: Changes in utilities and total fixed costs, seen in the final NPV at year 20.

The next sensitivity analysis performed was to see the effects of NPV when the utilities and total fixed costs fluctuated. The utility costs had the greatest affect, as seen with the steeper slope, while changes in the total fixed cost had little effect. Though, both parameters still obtained a very positive NPV when the costs fluctuated from 50% to 150% of their original value.

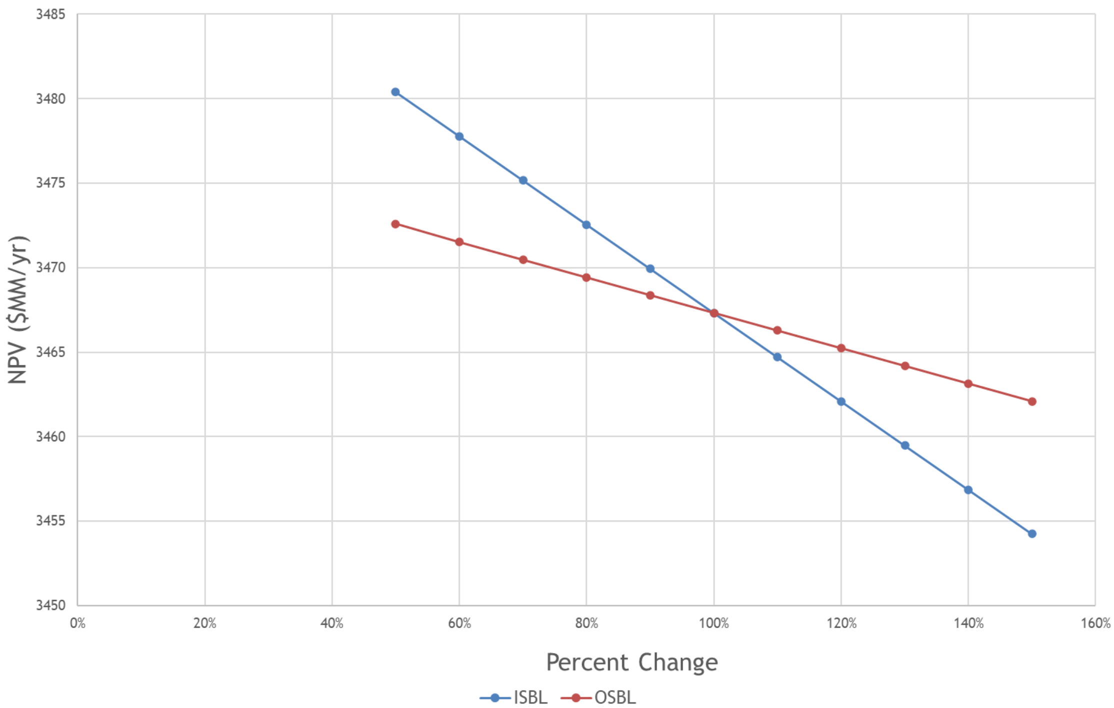


Figure 17: Changes in ISBL and OSBL, seen in the final NPV at year 20.

Contrary to the seemingly steep slopes, changes in ISBL and OSBL had little to no effect on the final NPV of the project. This is due to the high profits obtained in the project compared to the relatively low ISBL and OSBL costs. For example, the ISBL cost is only 7.15% of the gross profits, numerically speaking. Therefore, changes in ISBL and OSBL have little to no effect on the economics of the project. Though, the fluctuations in ISBL have a greater effect than that of OSBL because OSBL is only a percentage of ISBL.

Recommendations

Throughout the course of this project, and with much discussion from profession engineers, different issues were raised about the viability of the methylation of toluene process. Many considerations need to be continued further and many corrections could be made to improve the course of this endeavor. Included below are five different recommendations that could greatly improve the process feasibility and economics.

The first recommendation is to investigate a dividing wall column instead of the reactive distillation column T-204. One of the goals of this process was to do a heat exchanger network in order to save on utility costs and make the process more environmentally friendly. With a dividing wall column there would be less energy consumption, less overall capital investment on equipment, and lower overall maintenance costs.

The second recommendation is to further analyze the 5 different distillation columns in order to further lower the utilities. There could possibly be done with more design specifications that would give the same purity as needed for the process. This could include lowering the number of stages, lowering the pressure of the column, or lowering the reflux ratio of the column. All of these changes could potentially save money, making the process more economically viable.

The third recommendation is to perform a more in-depth economic comparison of the next best alternative. This would give insight into the economics of not just our process, but similar processes like UOP Honeywell's isomerization unit. This comparison would be beneficial to weigh the economics of the methylation of toluene against competing processes to see which process will be more profitable.

The last recommendation is to do an economic comparison for multiple reactor trains and distillation columns that could be linked in series. Due to the output of this process, 4.5 MMTPA, a series of reactors and/or distillation columns could reduce the size of the different pieces of equipment. This reduction in size could potentially reduce utilities costs as well as capital investments. It might be more feasible too during installation to have shorter columns and smaller reactors.

Appendices

A1.) - Design Basis

We propose building a world class paraxylene production plant in Dalian, China. We expect to produce 4.5 MMTPA of paraxylene with a purity 99.9%. With the implementation of the methylation of toluene to paraxylene, we are targeting a recovery of 99% of paraxylene from the reformed naphtha. We plan to have the construction of the plant done by 2025, with a 25% operating rate the first year, a 50% operating rate the next year, and then decide future operating rates based on the current year paraxylene demand.

Plant Capacity

The consumption rates of paraxylene globally are quickly increasing and are creating an opportunity to capitalize on the need for increased paraxylene production in the future. The largest producers of paraxylene are Hengli Petrochemical and Reliance Industries, which have facilities that produce 4.5 MMTPA and 4.3 MMTPA of PX, respectively [16,17]. To be competitive against the world's largest producers of PX, we need to the ability to match their production rates. Larger production rates also give room to decrease or increase production rates based on the current PX economy. A larger production facility will also create higher revenues that should help offset times of low PX prices. Therefore, we choose to build a world class 4.5 MMTPA PX facility.

Product Composition

Most PX production facilities target a purity of 99.5-99.7 wt% with a recovery greater than 98% [18]. The market is now demanding purities of PX that are 99.9 wt% and above [19]. This is due to the shift of higher quality PTA and its use in higher quality polyester fibers. For our PX plant, we will target a PX purity of 99.9 wt% and a recovery of greater than 99%. Increasing the recovery of PX from the reformed naphtha will increase the profit margin as more naphtha is being turned into PX instead of by-products. In our simulations we were able to achieve a paraxylene purity of 99.5% in distillation column T-205. Based on the bottom products flow out of T-205, 6,234.9 kg/hr of paraxylene and 31.32 kg/hr of toluene, the composition of our product stream is 99.5% paraxylene and 0.5% toluene.

The System of Units to be Used

The system of units that our plant will be design with will be the metric system. Our plant will be based in China, so for ease of communication with Chinese contractors, the metric system will be used.

Details of Raw Materials, Compositions, and their Availability

The raw materials that are needed in our proposed production of PX is naphtha and methanol. Currently, China is importing large amounts of naphtha with very little exports of naphtha [20]. In 2019 Zhejiang Petrochemical started up its naphtha cracker, which required 200,000 metric tons per month of naphtha [21]. This caused naphtha imports to rise in China with the additional consumption by Zhejiang Petrochemical. Also, China is currently the largest importer of methanol, with China accounting for 22% of worlds imports in 2018 [22]. China's large industrial sector consumes more naphtha and methanol then it can produce, leading it to import large amounts of the commodities. The addition of our world class PX production facility will ultimately increase the amount of naphtha and methanol being imported into the country. The feed into our process is a reformed naphtha feedstock with a total flow rate of 1,527,238 kg/hr and a composition of 27.8% toluene, 23.6% $C9^{+}$ aromatics, 15.2% m-xylene, 10.1% benzene, 8.9% o-xylene, 6.8% p-xylene, 6.6% ethylbenzene, and 1% propane. The methanol feed into the methylation of toluene process has a flow rate of 128,168 kg/hr and a composition of 100% methanol. The solvent feed into the reactive distillation column T-204 in the methylation of toluene process has flow rates of 152.28 kg/hr of DTBB and 53.7 kg/hr of TBB which gives a solvent composition of 74% DTBB and 26% TBB.

Location of Plant

The location of our proposed paraxylene production facility will be built in Dalian, China (See Figure 1). China's demand for paraxylene has been growing quickly over the past few decades that China is the largest manufacturer of polyester fibers in the world. Building our plant near the costumer will reduce transportation costs and support the heavy industrialization that is taking place in China. The town of Dalian, more specifically, is the prime location for our plant. In Dalian, the largest producers of PTA are located in Dalian. The Hengli Petrochemical plant and Yisheng Dahua Petrochemical Company are located in Dalian [16]. This gives our plant very close proximity to potentially very big costumers of PX. Another advantage of Dalian is being a peninsula, shipping routes can easily be implanted form the ports of Dalian to the major ports along China's trade routes. The introduction of a new PX production facility will require imports of methanol and naphtha, shipping of paraxylene and other by-products. Being located near the ports of Dalian will save on transportation costs for both import and exporting goods. Other advantages of choosing Dalian as location of our plant is the availability of many petrochemical infrastructure resources and low cost of utilities (see following Utilities section) [16].

***Figure A1-1*:** Location of our proposed paraxylene production facility. *Figures adapted taken from Google Maps ©.*

Utilities

The production of paraxylene requires electricity, fuel gas, and cooling water to power, cool, and heat the process equipment involved. In 2019, the price of electricity for business in China was \$0.097 USD per kWh [23]. The Chinese government has made it a top economic importance to make a unified nation grid system to improve the reliability of electricity in the country [24]. In Dalian, China, the current industrial infrastructure helps mitigates any electrical reliability risks, although a general risk of reliability still exists. In January 2020, the current price of natural gas, or fuel gas, was \$0.38 USD per m^3 of natural gas [25,26]. China currently imports large amounts of liquified natural gas (LNG) to meet its growing demand for natural gas. Consumption of natural gas is growing rapidly in China making it hard for suppliers to meet demand as China's energy grid is still in developing [26]. As industrialization in China continues to increase, LNG imports will have to increase in order for our PX plant to have a reliable supply of natural gas. An estimate for the cost of process water is to assume that it takes approximately 2 kWh for a recirculating cooling water system to circulate 1000 gal of water [27]. It is also estimated that it costs \$0.02/1000 gal of water to account for the cost of the water make-up and chemical treatments [27]. This gives us a cost of approximately \$0.214/1000 gal for process water.

A2.) – BFD & Material Balance

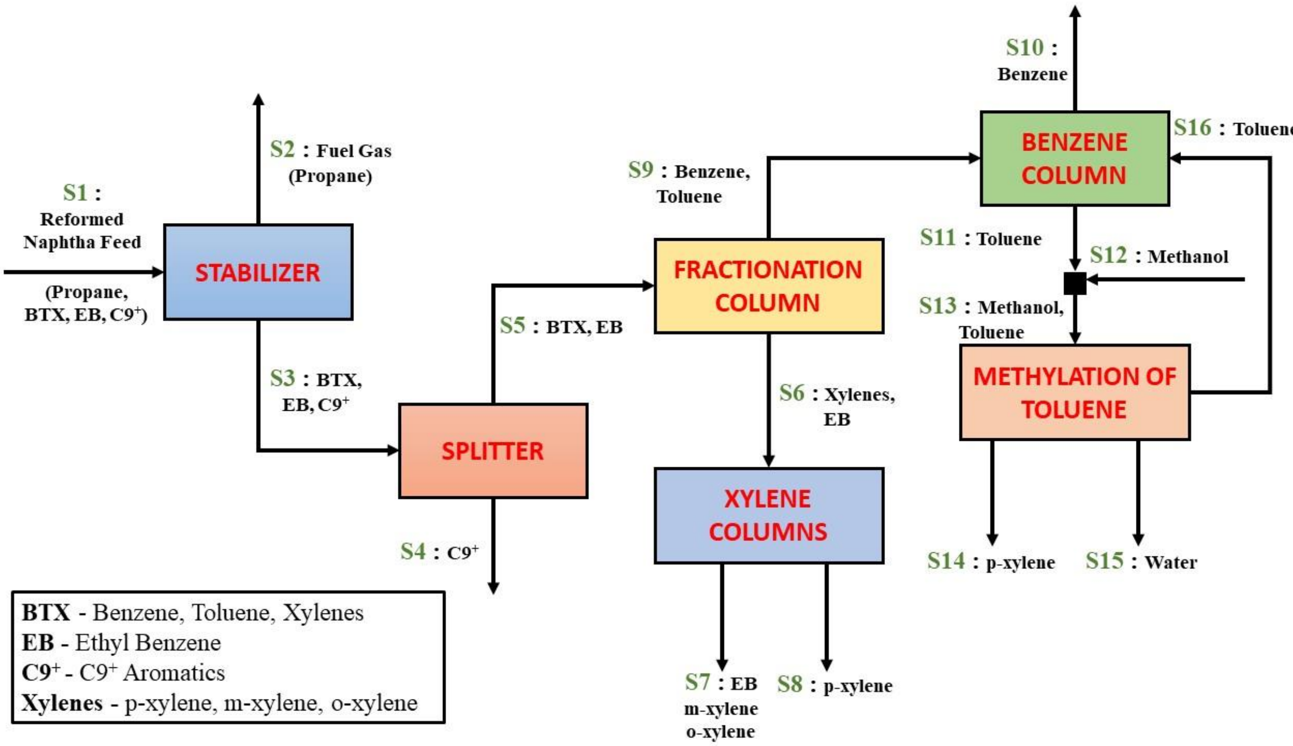


Figure A2-1: BFD of Para-xylene production through the methylation of toluene.

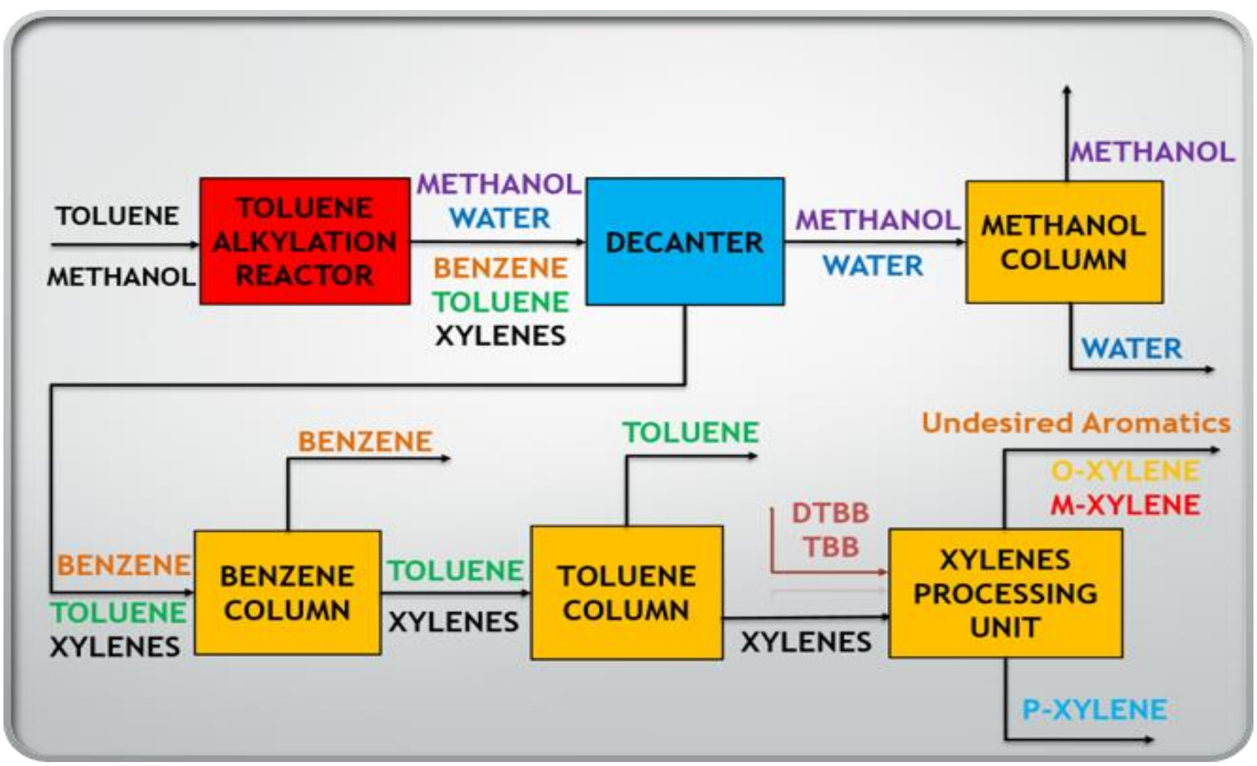


Figure A2-2: BFD of methylation of toluene unit.

Table A2-1: Material balance of the para-xylene production process through the methylation of toluene.

	S1 - Naphtha Reformate Feed		**S2** - Stabilzer (Top)		**S3** - Stabilzer (Bot)		**S5** - Splitter (Top)		**S4** - Splitter (Bot)		**S9** - Fractionation Column (Top)		**S6** - Fractionation Column (Bot)	
Species	**wt%**	**m (kg/hr)**	**wt%**	**m (kg/hr)**	**wt%**	**m (kg/hr)**	**wt%**	**m (kg/hr)**	**wt%**	**m (kg/hr)**	**wt%**	**m (kg/hr)**	**wt%**	**m (kg/hr)**
Propane	1.0%	15272	100.0%	15272	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0
Benzene	10.1%	154601	0.0%	0	10.2%	154601	13.4%	153055	0.4%	1546	26.4%	151525	0.3%	1531
Toluene	27.8%	424493	0.0%	0	28.1%	424493	36.7%	420248	1.2%	4245	72.6%	416046	0.7%	4202
ETHYLBENZENE	6.6%	100688	0.0%	0	6.7%	100688	8.7%	99681	0.3%	1007	0.2%	997	17.3%	98684
p-XYLENE	6.8%	104167	0.0%	0	6.9%	104167	9.0%	103126	0.3%	1042	0.2%	1031	17.9%	102094
m-XYLENE	15.2%	232427	0.0%	0	15.4%	232427	20.1%	230102	0.6%	2324	0.4%	2301	39.9%	227801
o-XYLENE	8.9%	135291	0.0%	0	8.9%	135291	11.7%	133938	0.4%	1353	0.2%	1339	23.2%	132599
C9+ (TMB)	23.6%	360299	0.0%	0	23.8%	360299	0.3%	3603	96.9%	356696	0.0%	36	0.6%	3567
SUM:	100.0%	1527238	0.0%	15272	100.0%	1511966	100.0%	1143754	100.0%	368212	100.0%	573275	100.0%	570479

	S10 - Benzene Column (Top)		**S11** - Benzene Column (Bot)		**S12/S13** - Methylation of Toluene (In)		**S15/S16** - Methylation of Toluene (Out)		**S14** - Methylation of Toluene (PX)		**S7** - Xylene Separation (EB,MX,OX)		**S8** - Xylene Separation (PX)	
Species	**wt%**	**m (kg/hr)**	**wt%**	**m (kg/hr)**	**wt%**	**m (kg/hr)**	**wt%**	**m (kg/hr)**	**wt%**	**m (kg/hr)**	**wt%**	**m (kg/hr)**	**wt%**	**m (kg/hr)**
Water	0.0%	0	0.0%	0	0.0%	0	78.2%	78153	0.2%	789	0.0%	0	0.0%	0
Methanol	0.0%	0	0.0%	0	25.5%	143226	2.8%	2836	0.0%	29	0.0%	0	0.0%	0
Benzene	97.3%	150009	0.4%	1515	0.3%	1515	1.5%	1500	0.0%	15	0.3%	1515	0.0%	15
Toluene	2.7%	4160	98.3%	411885	73.3%	411885	8.2%	8155	0.0%	82	0.9%	4160	0.0%	42
ETHYLBENZENE	0.0%	10	0.2%	987	0.2%	987	1.0%	977	0.0%	10	21.0%	97697	0.9%	987
p-XYLENE	0.0%	10	0.2%	1021	0.2%	1021	4.7%	4661	99.8%	461427	0.2%	1021	95.6%	101073
m-XYLENE	0.0%	23	0.5%	2278	0.4%	2278	2.3%	2255	0.0%	23	48.5%	225523	2.2%	2278
o-XYLENE	0.0%	13	0.3%	1326	0.2%	1326	1.3%	1313	0.0%	13	28.2%	131273	1.3%	1326
C9+ (TMB)	0.0%	0	0.0%	36	0.0%	36	0.0%	35	0.0%	0	0.8%	3531	0.0%	36
SUM:	100.0%	154227	100.0%	419048	100.0%	562274	100.0%	99885	100.0%	462388	100.0%	464721	100.0%	105757

A3.) – Finalized PFD

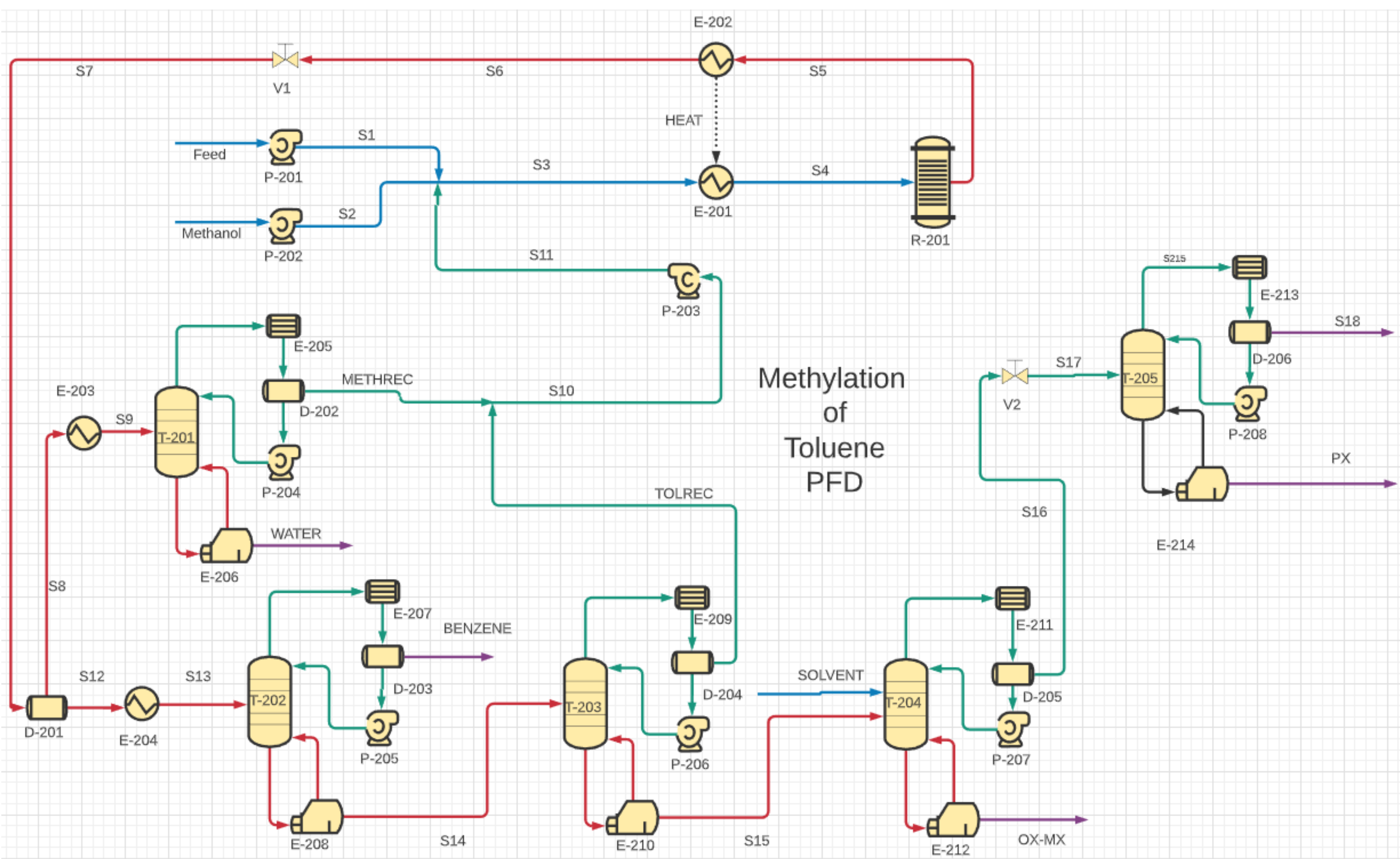


Figure A3-1: Finalized PFD of para-xylene production through the methylation of toluene

*All equipment is described in the equipment list.

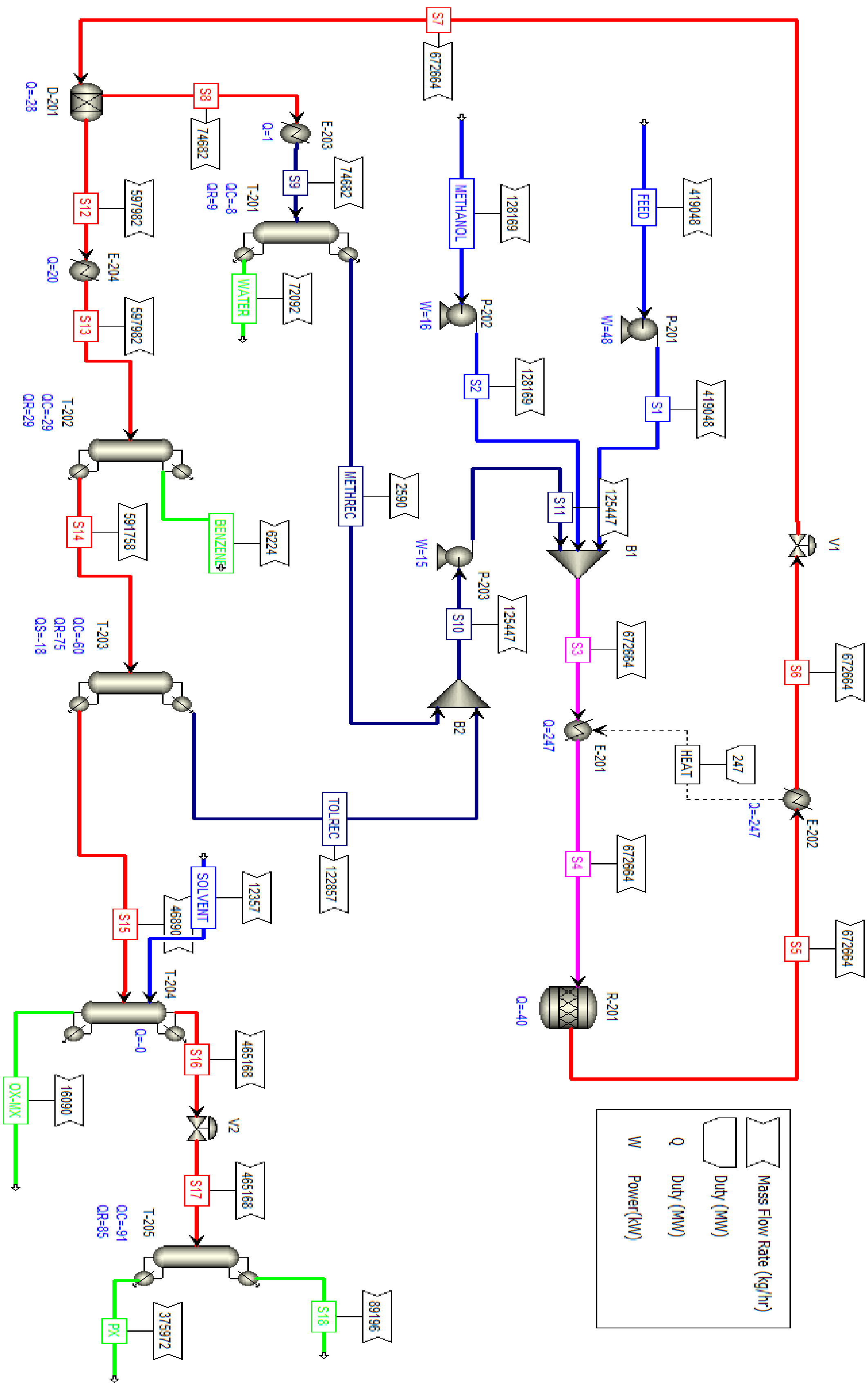


Figure A3-1: ASPEN screenshot of the methylation of toluene.

A4.) – Final Material Balance

Table A4-1: Material balance from the ASPEN simulation for the methylation of toluene, corresponding to the finalized PFD

Stream Name	Units (SI)	Feed	Methanol	S1	S2	S3	S4	S5	S6	S7	S8	S9	WATER	METHREC
Vapor Fraction		0.0	0.0	0.0	0.0	0.0	1.0	1.0	0.5	0.5	0.0	0.0	0.0	0.0
Temperature	C	25.0	25.0	25.1	25.1	20.1	444.4	525.0	120.0	95.7	95.7	101.8	106.5	64.2
Pressure	bar	1.0	1.0	4.0	4.0	4.0	4.0	4.0	4.0	2.0	2.0	1.2	1.2	1.0
Enthalpy	kJ/hr	582.5	-9651.8	584.2	-9651.2	-8977.7	-84.9	-1533.9	-10426.6	-10426.6	-11467.0	-11446.1	-11241.4	-195.2
Density	kg/cum	864.7	792.9	864.6	792.9	884.6	4.6	4.1	345.4	31.5	912.8	906.2	911.1	746.8
Total Mass Flow	Kg/hr	419048.0	128168.6	419048.0	128168.6	672663.8	672663.8	672663.8	672663.8	672663.8	74681.9	74681.9	72091.6	2590.3
Total Volumetric Flow	cum/hr	484.6	161.6	484.7	161.7	760.4	146746.0	163741.4	1947.4	21369.9	81.8	82.4	79.1	3.5
Component Mass Flow														
Methanol	kg/hr	0.0	128168.6	0.0	128168.6	130713.4	130713.4	2614.3	2614.3	2614.3	2614.3	2614.3	69.5	2544.7
Benzene	kg/hr	1676.2	0.0	1676.2	0.0	2450.5	2450.5	6979.6	6979.6	6979.6	0.0	0.0	0.0	0.0
Water	kg/hr	0.0	0.0	0.0	0.0	45.6	45.6	72067.6	72067.6	72067.6	72067.6	72067.6	72022.0	45.6
Toluene	kg/hr	412929.9	0.0	412929.9	0.0	534232.2	534232.2	155185.4	155185.4	155185.4	0.0	0.0	0.0	0.0
Ethyl Benzene	kg/hr	251.4	0.0	251.4	0.0	252.4	252.4	252.4	252.4	252.4	0.0	0.0	0.0	0.0
M-xylene	kg/hr	2095.2	0.0	2095.2	0.0	2098.3	2098.3	2162.9	2162.9	2162.9	0.0	0.0	0.0	0.0
P-xylene	kg/hr	838.1	0.0	838.1	0.0	1613.9	1613.9	432079.6	432079.6	432079.6	0.0	0.0	0.0	0.0
O-xylene	kg/hr	1257.1	0.0	1257.1	0.0	1257.4	1257.4	1322.0	1322.0	1322.0	0.0	0.0	0.0	0.0
C9+ Aromatics	kg/hr	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
DTBB	kg/hr	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
TBB	kg/hr	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
TBMX	kg/hr	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Stream Name	Units (SI)	S10	S11	S12	S13	BENZENE	S14	TOLREC	S15	SOLVENT	OX-MX	S16	S17	S18	PX
Vapor Fraction		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.5	0.0	0.0
Temperature	C	75.7	75.8	95.7	154.8	100.4	156.1	78.0	163.0	160.0	162.9	162.9	142.3	125.7	137.8
Pressure	bar	1.0	4.0	2.0	2.0	1.8	2.0	1.8	2.0	2.0	2.0	2.0	1.2	1.0	1.0
Enthalpy	kJ/hr	88.8	89.3	26.9	746.0	47.8	706.9	284.0	303.5	-79.1	-76.5	300.7	300.7	79.6	-51.7
Density	kg/cum	819.3	819.2	795.6	734.4	791.6	733.3	814.1	727.5	757.8	753.8	727.5	27.9	765.2	755.0
Total Mass Flow	Kg/hr	125447.1	125447.1	597981.9	597981.9	6224.2	591757.7	122856.8	468900.9	12357.1	16090.0	465167.9	465167.9	89195.6	375972.4
Total Volumetric Flow	cum/hr	153.1	153.1	751.6	814.3	7.9	807.0	150.9	644.6	16.3	21.3	639.4	16658.5	116.6	498.0
Component Mass Flow															
Methanol	kg/hr	2544.7	2544.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Benzene	kg/hr	774.3	774.3	6979.6	6979.6	6205.2	774.3	774.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Water	kg/hr	45.6	45.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Toluene	kg/hr	121302.3	121302.3	155185.4	155185.4	19.0	155166.4	121302.3	33864.1	0.0	0.0	33864.1	33864.1	31984.2	1879.9
Ethyl Benzene	kg/hr	1.0	1.0	252.4	252.4	0.0	252.4	1.0	251.4	0.0	251.4	0.0	0.0	0.0	0.0
M-xylene	kg/hr	3.1	3.1	2162.9	2162.9	0.0	2162.9	3.1	2159.8	0.0	2159.8	0.0	0.0	0.0	0.0
P-xylene	kg/hr	775.8	775.8	432079.6	432079.6	0.0	432079.6	775.8	431303.8	0.0	0.0	431303.8	431303.8	57211.3	374092.5
O-xylene	kg/hr	0.3	0.3	1322.0	1322.0	0.0	1322.0	0.3	1321.7	0.0	1321.7	0.0	0.0	0.0	0.0
C9+ Aromatics	kg/hr	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
DTBB	kg/hr	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	9135.8	9135.8	0.0	0.0	0.0	0.0
TBB	kg/hr	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	3221.3	3221.3	0.0	0.0	0.0	0.0
TBMX	kg/hr	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

*DTBB is 4,4'-Di-tert-Butylbiphenyl, TBB is 4,5,6,7-Tetrabromo-2-azabenzimidazole, and TBMX is tert-butyl m-xylene

A5.) – Equipment Sizing (From Aspen Plus Simulation) and Corresponding Description List

Table A5-1A: Equipment sizing for the methylation of toluene process, along with a detailed description of the equipment's function

Unit	Equipment Type	Size/Power	Material	Description
P-201	Centrifugal Pump	Fluid head:35m W:48kW	Carbon Steel	P-201 increases the pressure of the feed to 4 bar and pumps the feed to S1.
P-202	Centrifugal Pump	Fluid head:39m W:16 kW	Carbon Steel	P-202 increases the pressure of the methanol feed to 4 bar and pumps the methanol to S2.
P-203	Centrifugal Pump	Fluid head:37m W:15 kW	Carbon Steel	P-203 increases the pressure of the recycle stream to 4 bar.
R-201	Packed Bed Reactor	V:141180L D:7m L:4m(tangent/tangent)	Stainless Steel	This is a packed bed alkylation reactor that converts toluene into paraxylene at a pressure of 4 bar and a temperature of 977 ^{o}F.
D-201	Decanter	V:88805L D:3m L:10m(tangent/tangent)	Carbon Steel	D-201 separates the aqueous and organic phases of S7. S7 contains a mixture of water and immiscible non-polar organic compounds such as toluene and xylene.
T-201-tower	Distillation Column	D:11m L:22m Sieve trays:25	Carbon Steel	The rich water stream from Decanter D-201 contains a large amount of methanol. This methanol will be separated and recycled back to the reactor. The distillation column T-201 is used for this purpose.
T-202-tower	Distillation Column	D:18.5m L:37m Sieve trays:50	Carbon Steel	T-202 separates benzene from a stream rich in toluene and xylenes.
T-203-tower	Distillation Column	D:15.5m L:31m Sieve trays:40	Carbon Steel	T-203 separates toluene from a stream rich in xylenes.
T-204-tower (Sep2)	eactive Distillation Colun	V:77237L D:3m L:10m(tangent/tangent)	Carbon Steel	T-204 reacts DTBB and TBB solvents with m-xylene to produce a tert-butyl-m-xylene which has a different boiling point than p-xylene which allows it to be separated out in the column.
T-205-tower	Distillation Column	D:9m L:18m Sieve trays:19	Carbon Steel	T-205 separates p-xylene (our product) from a stream rich in toluene and some benzene.
D-202	Horizontal Reflux Drum	V:4270L D:1.2 L:4m(tangent/tangent)	Carbon Steel	A temporary drum that collects the condensate from the condensor E-205.

Table A5-1B: Continued equipment sizing for the methylation of toluene process, along with a detailed description of the equipment's function

D-203	Horizontal Reflux Drum	V:36296L D:2.4 L:7.8m(tangent/tangent)	Carbon Steel	A temporary drum that collects the condensate from the condensor E-207.
D-204	Horizontal Reflux Drum	V:82141L D:3m L:10m(tangent/tangent)	Carbon Steel	A temporary drum that collects the condensate from the condensor E-209.
D-205	Horizontal Reflux Drum	N/A	Carbon Steel	A temporary drum that collects the condensate from the condensor E-211.
D-206	Horizontal Reflux Drum	V:123299L D:3.7m L:11.7m(tangent/tangent)	Carbon Steel	A temporary drum that collects the condensate from the condensor E-213.
P-204	Centrifugal Reflux Pump	Fluid head:25m W:11kW	Carbon Steel	Draws the liquid condensate from the reflux drum D-202 and sends it back to the top tray of T-201 as a reflux.
P-205	Centrifugal Reflux Pump	Fluid head:28m W:16 kW	Carbon Steel	Draws the liquid condensate from the reflux drum D-203 and sends it back to the top tray of T-202 as a reflux.
P-206	Centrifugal Reflux Pump	Fluid head:29m W:12 kW	Carbon Steel	Draws the liquid condensate from the reflux drum D-204 and sends it back to the top tray of T-203 as a reflux.
P-207	Centrifugal Reflux Pump	N/A	Carbon Steel	Draws the liquid condensate from the reflux drum D-205 and sends it back to the top tray of T-204 as a reflux.
P-208	Centrifugal Reflux Pump	Fluid head:31 W:23kW	Carbon Steel	Draws the liquid condensate from the reflux drum D-206 and sends it back to the top tray of T-205 as a reflux.

Table A5-1C: Continued equipment sizing for the methylation of toluene process, along with a detailed description of the equipment's function

E-205	TEMA Shell and Tube HE	Area:527sqm Q: -8.362MW	Stainless Steel	Condenses the vapor distillate in the column T-201 to a liquid.
E-207	TEMA Shell and Tube HE	Area:634sqm Q: -28.722MW	Stainless Steel	Condenses the vapor distillate in the column T-202 to a liquid.
E-209	TEMA Shell and Tube HE	Area:1073sqm Q: -59.592MW	Stainless Steel	Condenses the vapor distillate in the column T-203 to a liquid.
E-211	TEMA Shell and Tube HE	N/A	Stainless Steel	Condenses the vapor distillate in the column T-204 to a liquid.
E-213	TEMA Shell and Tube HE	Area:1290sqm Q: -87.247MW	Stainless Steel	Condenses the vapor distillate in the column T-205 to a liquid.
E-206	Kettle Reboiler	Area:431sqm Q: 8.6MW	Carbon Steel	The reboiler heats the liquid in the bottom of the distillation column T-201 to 106.5 ^{o}C and provides a heat souce for the operating temperature of the column.
E-208	Kettle Reboiler	Area:1975sqm Q: 29MW	Carbon Steel	The reboiler heats the liquid in the bottom of the distillation column T-202 to 156.1 ^{o}C and provides a heat souce for the operating temperature of the column.
E-210	Kettle Reboiler	Area:7178sqm Q: 75MW	Carbon Steel	The reboiler heats the liquid in the bottom of the distillation column T-203 to 163.0 ^{o}C and provides a heat souce for the operating temperature of the column.
E-212	Kettle Reboiler	N/A	Carbon Steel	The reboiler heats the liquid in the bottom of the distillation column T-204 to 162.9 ^{o}C and provides a heat souce for the operating temperature of the column.
E-214	Kettle Reboiler	Area:3155sqm Q: 80MW	Carbon Steel	The reboiler heats the liquid in the bottom of the distillation column T-205 to 144.8 ^{o}C and provides a heat souce for the operating temperature of the column.
E-201	TEMA Shell and Tube HE	Area:1392 sqm Q: 247MW	Stainless Steel	Increases the temperature of the reactor feed to 445 ^{o}C.
E-202	TEMA Shell and Tube HE	Area:1392 sqm Q: -247MW	Stainless Steel	Cools down the reactor effluent stream S5 to 120 ^{o}F.
E-203	TEMA Shell and Tube HE	Area:18sqm Q: 1MW	Carbon Steel	Increases the temperature of S8 to 102 ^{o}C.
E-204	TEMA Shell and Tube HE	Area:825 sqm Q: 20MW	Carbon Steel	Increases the temperature of S12 to 150 ^{o}C.

A6.) – Distillation Sizing Calculation and Discussion

Distillation Column Sizing

The methanol column (block T-201 in the APSEN simulation) in our process separates methanol from water and recycles it back to the reactor. The sizing of this column will be performed using the general method described below:

1. The number of stages required for the separation of methanol from water will be determined by first specifying the purity of methanol we want to be recycled back to the reactor. Since the methanol stream is a recycle stream, a purity and recovery of 99 mol% of methanol is sufficient and economic.
2. Once the desired purity and recovery of the methanol is determined, the Fenske equation can be used to determine the minimum number of stages at total reflux required to perform the desired separation.
3. Next, determine the efficiency of the trays by using the O'Connell equation.
4. The number of required trays in the column will be the minimum number of stages divided by the tray efficiency.
5. The height of the column is calculated by multiplying the number of trays by the tray spacing and then adding extra height to account for the dome and sump of the column.
6. The diameter of the column is calculated using the "Heat Factor" rule of thumb to give an approximate result based on the reboiler duty.

Number of Stages

The Fenske equation is an equation used for calculating the minimum number of theoretical plates required to separate a binary feed stream by a continuous fractional distillation column that is at total reflux [28].

$$N = \frac{\log\left[\left(\frac{X_d}{1-X_d}\right)\left(\frac{1-X_b}{X_b}\right)\right]}{\log \alpha_{avg}}$$

Where N is the minimum number of theoretical plates required at total reflux, X_d and X_b are the mole fractions of the more volatile component in the distillate and bottoms of the column, respectively, and α_{avg} is the average relative volatility of the more volatile component to the less volatile component [28]. The α_{avg} is defined by the following equation:

$$\alpha_{avg} = \sqrt{\alpha_{top}\alpha_{bot}}$$

Where α_{top} and α_{bot} are the relative volatilities of light key to heavy key at top and bottom of the column, respectively. α is defined as the ratio of the K values for component 1 and component 2, or $\alpha = K_{MEOH}/K_{WATER}$. The K values for the distillate and bottoms streams were calculated by using the KVL property set in ASPEN.

Top of Column: $K_{MEOH} = 1.02$ $K_{WATER} = 0.36$

Bottom of Column: $K_{MEOH} = 10.79$ $K_{WATER} = 0.995$

$\alpha_{top} = K_{MEOH,top}/K_{WATER,top} = 1.02/0.36 = 2.833$

$\alpha_{bot} = K_{MEOH,bot}/K_{WATER,bot} = 10.79/0.995 = 10.844$

$$\boldsymbol{\alpha_{avg} = \sqrt{\alpha_{top}\alpha_{bot}} = \sqrt{2.833 * 10.85} = 5.543}$$

The desired purity of methanol being recycled back is 99 mol% and since we require a 99 mol% recovery of methanol in the distillate, methanol in the bottoms will be 1 mol%. Therefore, the theoretical number of minimum stages is calculated below:

$$\boldsymbol{N = \frac{\log\left[\left(\frac{0.99}{1-0.99}\right)\left(\frac{1-0.01}{0.01}\right)\right]}{\log 5.543} = 5.366}$$

The minimum number of theoretical plates required at total reflux is 5.366. In order to calculate the actual number of trays needed for the methanol column, we will have to divide the minimum number of theoretical plates by the plate efficiency. The plate efficiency can be calculated by using the O'Connell equation [29]:

$$\boldsymbol{\eta = 50.3\left(\alpha_{avg} * \mu\right)^{-0.226}}$$

Where μ is the viscosity of the feed to the column and α_{avg} was defined previously above. The viscosity of the feed was calculated using the property set MUMX in ASPEN.

Viscosity of Feed: $\mu = 0.274$ cp

$$\boldsymbol{\eta = 50.3(3.331 * 0.274)^{-0.226} = 51.35\%}$$

$$\boldsymbol{N_{actual} = N/\eta = 5.543/0.5135 = 10.795 \approx 11}$$

The number of actual trays required for the methanol/water column to achieve a 99 mol% purity of methanol in the distillate is 11 trays.

Height of the column

We will assume that the distance between each sieve tray will be 2 ft, enough room for a person to perform maintenance on the column. To calculate the total height of the methanol column we have to sum up all the heights added by the sieve trays and add a certain percentage more to the height to account for the dome and sump of the distillation column. We will assume that the height added by the sump and dome of the column will be 15% of the height of the trays. The formula that we will use to calculate the height of the column is as follows:

$$\boldsymbol{H = (N_{actual} - 1) * \phi * 1.15}$$

Where φ is tray spacing and (N_{actual} – 1) accounts for the last tray, meaning it does not count towards the total height of the column. The height of the column was calculated to be 23 ft tall.

$$\boldsymbol{H = (11 - 1) * 2 * 1.15 = 23} \text{ ft}$$

Diameter of the Column

The diameter of the column was determined by using the "Heat Factor" rule of thumb [30]. The correlation states the following:

$$\boldsymbol{Q/d^2 = 350,000}$$

Where Q is the duty of the reboiler in BTU/hr and d is the diameter of the column in ft. The diameter of the column was calculated to be 9.17 ft.

$$\boldsymbol{d = \sqrt{29430551\frac{BTU}{hr}/350,000} = 9.17\ ft}$$

Comparison with APSEN results

Table A6-1: Tabulated results of the hand calculated sizing of the methanol column and the ASPEN calculated results.

Parameters	Hand Calculated	ASPEN v8.4
Number of Trays	11	19
Height of the Column (ft)	32.2	41.4
Diameter of the Column (ft)	9.17	5.64

The number of trays calculated by hand for the methanol column is noticeably different from what was calculated by ASPEN. This is the case because the Fenske and O'Connell equations use correlations to give an approximate result. Both the Fenske and O'Connell equation use average volatility to describe the volatility of the column, which does not accurately describe the stage to stage changes in volatility. Similarly, the O'Connell equation uses the viscosity of the feed which does not take in account the stage to stage change in the viscosity. ASPEN calculates the stage to stage vapor-liquid equilibrium and accounts for the temperature profile, viscosity changes, and the volatility changes in the column. These stage to stage calculations allow ASPEN to use rigorous VLE calculations to provide the most accurate representation of the liquid and vapor compositions at each tray. Therefore, number of stages that ASPEN says the column needs will be the most accurate representation of what is needed in the column to perform the separation with the desired specifications. The heights of the columns calculated by hand and ASPEN are significantly different mainly due to the difference in the calculated number of trays. To determine the height of the column, both ASPEN and the hand calculations assumed the tray spacing was 2 ft. The hand calculations assumed that the sump and dome were 15% of the total stacked tray height where as APSEN most likely used some proprietary calculation within its software to determine total height. Again, there is a major difference in the hand calculated column diameter and the APSEN calculated column diameter. In the hand calculations, a rule of thumb was used to determine the diameter of the column based on the reboiler duty. This calculation assumes that the diameter of the column is dependent only on a constant value and the duty of the reboiler. This is a very rough estimation of the diameter of the column as it doesn't use thermodynamic correlations or include the type of trays in the column to determine optimal diameter. APSEN, on the other hand, calculates the vapor velocity through the column with sieve trays and determines the optimal diameter to prevent flooding/drying of the column.

Reboiler & Condenser Sizing

The size, or the heat transfer area required for the reboiler/condenser in the methanol column can be calculated with the following equation:

$$\boldsymbol{Q = F_T * A * U * \Delta T_{ln}} \quad \Rightarrow \quad \boldsymbol{A = Q/(F_T * U * \Delta T_{ln})}$$

Where Q is the heat duty of the reboiler/condenser, F_T is the F factor for the heat exchanger, A is the heat transfer area, U is the heat transfer coefficient, and ΔT_{ln} is the logarithmic mean of the stream temperatures going into the reboiler.

From ASPEN, we know the inlet/outlet temperatures of the bottoms and steam streams as well as the reboiler duty. We also assume that cooling water (CW) comes from the cooling tower at 86 ^oF and can reach a temperature of 122 ^oF.

$T_{steam,\ in}$ = 257 ^oF $T_{bot,\ out}$ = 223.63 ^oF ΔT_1 = 33.37 ^oF
$T_{steam,\ out}$ = 255.2 ^oF $T_{bot,\ in}$ = 222.75 ^oF ΔT_2 = 32.45 ^oF

$Q_{reboiler}$ = 29430551 BTU/hr

$T_{cw,\ in}$ = 86 ^oF $T_{top,\ out}$ = 147.61 ^oF ΔT_1 = 61.61 ^oF
$T_{cw,\ out}$ = 122 ^oF $T_{dis,\ in}$ = 149.70 ^oF ΔT_2 = 27.7 ^oF

$Q_{condenser}$ = -28534561 BTU/hr

We will assume that the reboiler/condenser will have pure counter current flow which will make the $F_T = 1$. The heat transfer coefficient between steam and water ($U_{steam/water}$) was found to range between 250-750 BTU/hr*ft^2*F and between water and an organic solvent to range from 50-150 BTU/hr*ft^2*F [31]. We will take an average value of these heat transfers coefficients, 500 BTU/hr*ft^2*F and 100 BTU/hr*ft^2*F, respectively, to use in the sizing calculations. The calculations of the reboiler/condenser size were carried out below:

$$\Delta T_{ln} = (\Delta T_1 – \Delta T_2) / \ln(\Delta T_1/\Delta T_2) = (33.37\text{-}32.45)/\ln(33.37/32.45) = 32.9\ ^oF$$

$$\boldsymbol{A = 29430551\frac{BTU}{hr}\Big/\left(1 * 500\frac{BTU}{hr * ft^2 * F} * 32.9\ F\right) = 1789\ ft^2}$$

The heat transfer area needed for the reboiler using steam as a utility is 1789 ft^2.

$$\Delta T_{ln} = (\Delta T_1 – \Delta T_2) / \ln(\Delta T_1/\Delta T_2) = (61.61\text{-}27.7)/\ln(61.61/27.7) = 42.42\ ^oF$$

$$\boldsymbol{A = 28534561\frac{BTU}{hr}\Big/\left(1 * 100\frac{BTU}{hr * ft^2 * F} * 42.42\ F\right) = 6726\ ft^2}$$

The heat transfer area needed for the condenser using cooling water as a utility is 6726 ft^2.

Pump Sizing

The size, or the power requirements, for the reflux pump can be calculated with the following equation:

$$\boldsymbol{Power = \rho g h F}$$

Where Power is the power of the pump in W, ρ is the density of the fluid in kg/m^3, g is the acceleration due to gravity (9.82 m/s^2), h is the height the pump needs to pump the fluid in m, and F is the flow rate of liquid through the pump in m^3/s.

In ASPEN, we created a pseudo-stream at stage 1 to get the density and flow rate of the reflux stream:

$$\rho_{reflux} = 746.84\ kg/m^3$$
$$F = 0.0084195\ m^3/s$$

We will assume that the reflux pump sits on the ground and pumps the reflux back up to the top of the distillation column. Using ASPEN's sizing of column (41.4 ft) and subtracting a few feet to discount the dome height, we will take h = 39 ft (11.89 m). The calculations of the pump size were carried out below:

$$\begin{aligned}\boldsymbol{Power} &= \mathbf{746.84}\frac{\boldsymbol{kg}}{\boldsymbol{m^3}} * \mathbf{9.82}\frac{\boldsymbol{m}}{\boldsymbol{s^2}} * \mathbf{11.89}\,\boldsymbol{m} * \mathbf{0.0084195}\frac{\boldsymbol{m^3}}{\boldsymbol{s}} \\ &= \mathbf{734.18}\,\boldsymbol{W}\end{aligned}$$

The power needed for the reflux pump is 734.18 W or 0.98 HP.

<u>Reflux Drum Sizing</u>

The size of the reflux drum will be calculated by assuming a 5 min residence time of the liquid at half volume and a length to diameter (L/D) ratio of 2. The following equation is used to find the volume of the reflux drum:

$$\boldsymbol{V = \tau * F} \quad \rightarrow \quad \boldsymbol{V_{vessel} = 2 * V} \quad \rightarrow \quad \boldsymbol{V_{vessel} = 2 * \tau * F}$$

Where V is the volume of half the reflux drum in m^3, V_{vessel} is the volume of the reflux drum in m^3, τ is the residence time in minutes, and F is the volumetric flow rate in m^3/min. The flow rate of the distillate was calculated in ASPEN and was determined to be Q = 0.50517 m^3/min. The volume of the reflux drum is calculated below:

$$\boldsymbol{V_{vessel} = 2 * 5\,min * 0.50517\,\frac{m^3}{min} = 5.0517\,m^3}$$

The diameter of the reflux drum can be calculated using the following equation using the assumption that the L/D ratio is 2.

$$\boldsymbol{V_{vessel} = \frac{\pi * D^2}{4} * L}$$

$$\boldsymbol{V_{vessel} = \frac{\pi * D^2}{4} * 2 * D}$$

$$D = \left(\frac{2 * V_{vessel}}{\pi}\right)^{1/3}$$

The calculations of the reflux drum size were carried out below:

$$D = \left(\frac{2 * 5.0517m^3}{\pi}\right)^{1/3} = 1.476\ m$$

$$L = 2 * D = 2 * 1.476\ m = 2.952\ m$$

The size of the reflux drum is a diameter of 1.476 m (4.84 ft), a length of 2.952 m (9.68 ft), and a total volume of 5.05 m^3 (178.4 ft^3).

A7.) – Utilities

Table A7-1A: Utility requirements for all the equipment in the methylation of toluene process

Utility Resources					
Name	**Equipment**	**Type**	**Usage**	**Units**	**Description**
E-203	Shell & Tube Heat Exchanger	Low Pressure Steam	2,104	lb/hr	Used for heating up 164,645 lb/hr of Stream S8 from 204.2 ^oF to 215.2 ^oF
E-206	Kettle Reboiler	Low Pressure Steam	31,231	lb/hr	Used for heating up the column T-201 and heats up 158,934 lb/hr of Stream WATER from 222.7 ^oF to 223.6 ^oF
E-205	Shell & Tube Heat Exchanger	Cooling Water	795,299	lb/hr	Used for condensing the distillate from column T-201 and cools 5,710 lb/hr of Stream METHREC from 149.7 ^oF to 147.6 ^oF
E-207	Shell & Tube Heat Exchanger	Cooling Water	2,731,529	lb/hr	Used for condensing the distillate from column T-202 and cools 13,722 lb/hr of Stream BENZENE from 213.0 ^oF to 212.7 ^oF
E-209	Shell & Tube Heat Exchanger	Cooling Water	7,411,310	lb/hr	Used for condensing the distillate from column T-203 and cools 1,033,750 lb/hr of Stream TOLREC from 134.2 ^oF to 133.7 ^oF
E-211*	Shell & Tube Heat Exchanger	Cooling Water	15,045,407	lb/hr	Used for condensing the distillate from column T-204 and cools 37,332 lb/hr of Stream S16 from 264.7 ^oF to 264.2 ^oF
E-213	Shell & Tube Heat Exchanger	Cooling Water	8,674,735	lb/hr	Used for condensing the distillate from column T-205 and cools 196,642 lb/hr of Stream S18 from 266.7 ^oF to 258.2 ^oF
R-201	Cooling Jacket	Cooling Water	3,827,650	lb/hr	Used for cooling the reactor R-201 to maintain a reactor temperature of 977 ^oF
E-204	Shell & Tube Heat Exchanger	Medium Pressure Steam	78,592	lb/hr	Used for heating up 1,318,320 lb/hr of Stream S12 from 204.2 ^oF to 310.7 ^oF
E-208	Kettle Reboiler	Medium Pressure Steam	113,952	lb/hr	Used for heating up the column T-202 and heats up 1,304,600 lb/hr of Stream S14 from 154.4 ^oF to 156.1 ^oF

Table A7-1B: Continued utility requirements for all the equipment in the methylation of toluene process

E-210	Kettle Reboiler	Medium Pressure Steam	293,578	lb/hr	Used for heating up the column T-203 and heats up 1,033,750 lb/hr of Stream S15 from 161.1 ^{o}F to 163.0 ^{o}F
E-212*	Kettle Reboiler	Medium Pressure Steam	154,897	lb/hr	Used for heating up the column T-204 and heats up 997,502 lb/hr of Stream OX-MX from 310.0 ^{o}F to 311.1 ^{o}F
E-214	Kettle Reboiler	Medium Pressure Steam	335,070	lb/hr	Used for heating up the column T-205 and heats up 828,877 lb/hr of Stream PX from 291.5 ^{o}F to 292.6 ^{o}F
P-201	Centrifugal Pump	Electricity	47.5	kW	Used for increasing the pressure of Stream Feed to 4 bar.
P-202	Centrifugal Pump	Electricity	15.8	kW	Used for increasing the pressure of Stream Methanol to 4 bar.
P-203	Centrifugal Pump	Electricity	15.0	kW	Used for increasing the pressure of Stream S11 to 4 bar before being combined with Streams FEED and METHANOL.
P-204	Centrifugal Pump	Electricity	0.7	kW	Reflux pump, used for pumping the reflux back to the top tray in column T-201.
P-205	Centrifugal Pump	Electricity	16.9	kW	Reflux pump, used for pumping the reflux back to the top tray in column T-202.
P-206	Centrifugal Pump	Electricity	25.1	kW	Reflux pump, used for pumping the reflux back to the top tray in column T-203.
P-207*	Centrifugal Pump	Electricity	22.8	kW	Reflux pump, used for pumping the reflux back to the top tray in column T-204.
P-208	Centrifugal Pump	Electricity	20.5	kW	Reflux pump, used for pumping the reflux back to the top tray in column T-205.

*Note: The sizing of E-211, E-212, P-207 was done using a previous simulation to get an estimated size of the column and power requirements for the reflux pump. These values are estimated to our best ability as we were not able to get a realistic reactive distillation column.

Table A7-1C: Overall calculated utility requirements for all equipment in the methylation of toluene process

Utilities							
Mode	Type	Rate	Units	Cost	Units	Cost/hr	$MM/year
Electricity	Power	694	KW-hr	0.084	$/kW	58.296	0.5
Cooling water	Water	10.23	MMGal/hr	120.5	$/MMGal	1,232.715	10
LP Steam	Steam	6,874	klb/hr	0.044	$/klb	300.06	2
HP Steam	Steam	199,394	klb/hr	0.048	$/klb	10,737.1	86
						Total	98 $MM/year

A8.) – Economic Analysis Calculations (Excel)

Equipment Class (Add / Remove)	ASPEN	crip	Te	Mat eria	C	Diam et	# Tr	S urf	P ow	Volu metric	Den sit	Purchased Cost ($MM)	Installed Cost	Interest Rate	Life (yr)	$MM/yr
Distillation Columns												$5.91	$23.63			
Pressure Vessels and Drums												$0.27	$1.10			
Heat Transfer Equipment												$3.52	$12.30			
Pumps and Drivers												$0.21	$0.82			
Turbines												$0.00	$0.00			
Compressors												$0.00	$0.00			
Reactors												$0.11	$0.42			
Reactor Vessel	-201	Carbon Ste	141									$0.11	$0.42	15%	3.00	$0.19
Solids Equipment												$0.00	$0.00			
TOTAL EQUIPMENT COST												$10.01	$38.28			$16.77

Utility Class (Add / Remove)	ASPE	script	Tem	Tem	Q Va	Fea	Units of Usa	Usa	Price of Utility	Plant Usage (U /year or Ehr /year) year	Cost of Utility ($MM/yr) $MM	$/unit product
Steam											$76.42	
Hot Oil or Fuel Gas											$0.00	
Cooling Water											$3.02	
Electric											$0.11	
Refrigeration											$0.00	
TOTAL UTILITY COST											$79.55	$0.00

Figure A8-1: Equipment cost calculations from excel.

ECONOMIC ANALYSIS

Average cash flow	$828.38 $MM/yr	NPV	10 years	2530.0 $MM	IRR	10 years	372.9%
Simple pay-back period	$ 0.10 yrs		15 years	3161.7 $MM		15 years	372.9%
Return on investment (10 yrs)	$9.63		20 years	3467.3 $MM		20 years	372.9%
Return on investment (15 yrs)	$10.74	NPV to yr	6	1563.4 $MM			

Figure A8-2: Economic indicator cost calculations from excel.

NPV

REVENUES AND PRODUCTION COSTS

	$MM/yr
Main product revenue	$3,158.16
Byproduct revenue	$42.13
Raw materials cost	$2,019.68
Utilities cost	$79.55
Consumables cost	$2.58
VCOP	$2,059.67
Salary and overheads	$4.39
Maintenance	$2.35
Interest	$0.00
Royalties	$0.00
FCOP	$6.75

CAPITAL COSTS

	$MM/yr
ISBL Capital Cost	$38.28
OSBL Capital Cost	$15.31
Engineering Costs	$7.66
Contingency	$11.48
Working Capital	$5.74
Total Fixed Capital Cost	$78.48

CONSTRUCTION SCHEDULE

Year	% FC	% WC	% FCOP	% VCOP
1	30%	0%	0%	0%
2	70%	0%	0%	0%
3	0%	100%	100%	50%
4	0%	0%	100%	100%
5	0%	0%	100%	100%
6	0%	0%	100%	100%
7	0%	0%	100%	100%
8	0%	0%	100%	100%
9	0%	0%	100%	100%
10	0%	0%	100%	100%
11+	0%	0%	100%	100%

ECONOMIC ASSUMPTIONS

Cost of Equity	25%	Debt ratio	55%	Depreciation Method	MACRS
Cost of Debt (interest rate)	8%	Tax Rate	25%	Depreciation Period	7 years
Cost of Capital	15.65%			Construction Period	2 years

Assign Percentages

CASH FLOW ANALYSIS - NPV

Project year	Cap Ex	Revenue	CCOP	Gr. Profit	Deprcn	Taxbl Inc	Tax Paid	Cash Flow	PV of CF	NPV
1	23.5	0.0	0.0	0.00	0.00	0.0	0.0	-23.5	-20.4	-20.4
2	54.9	0.0	0.0	0.00	0.00	0.0	0.0	-54.9	-41.1	-61.4
3	5.7	1579.1	1036.6	542.50	11.21	531.3	0.0	536.8	347.0	285.6
4	0.0	3158.2	2066.4	1091.75	19.22	1072.5	132.8	958.9	536.0	821.6
5	0.0	3158.2	2066.4	1091.75	13.73	1078.0	268.1	823.6	398.1	1219.7
6	0.0	3158.2	2066.4	1091.75	9.80	1081.9	269.5	822.2	343.7	1563.4
7	0.0	3158.2	2066.4	1091.75	7.01	1084.7	270.5	821.3	296.8	1860.2
8	0.0	3158.2	2066.4	1091.75	7.00	1084.7	271.2	820.6	256.4	2116.6
9	0.0	3158.2	2066.4	1091.7	7.01	1084.7	271.2	820.6	221.7	2338.3
10	0.0	3158.2	2066.4	1091.7	3.50	1088.2	271.2	820.6	191.7	2530.0
11	0.0	3158.2	2066.4	1091.7	0.00	1091.7	272.1	819.7	165.6	2695.6
12	0.0	3158.2	2066.4	1091.7	0.00	1091.7	272.9	818.8	143.0	2838.7
13	0.0	3158.2	2066.4	1091.7	0.00	1091.7	272.9	818.8	123.7	2962.3
14	0.0	3158.2	2066.4	1091.7	0.00	1091.7	272.9	818.8	106.9	3069.3
15	0.0	3158.2	2066.4	1091.7	0.00	1091.7	272.9	818.8	92.5	3161.7
16	0.0	3158.2	2066.4	1091.7	0.00	1091.7	272.9	818.8	80.0	3241.7
17	0.0	3158.2	2066.4	1091.7	0.00	1091.7	272.9	818.8	69.1	3310.8
18	0.0	3158.2	2066.4	1091.7	0.00	1091.7	272.9	818.8	59.8	3370.6
19	0.0	3158.2	2066.4	1091.7	0.00	1091.7	272.9	818.8	51.7	3422.3
20	-5.7	3158.2	2066.4	1091.7	0.00	1091.7	272.9	824.6	45.0	3467.3

Figure A8-3: NPV tables from excel.

REVENUES AND RAW MATERIAL COSTS

MASS BALANCE:		Mass Balance Closure	70%					
Process Stream Classes Add Remove	ASPEN Stream	Description	Units	Units/hr hr	Units/Unit MainProduct	Price $/unit	$MM/yr $MM	$/unit main product
Key Products								
Main Product		PX	kg	375972	1	$1.05	$3,158.16	$0.00
Total Key Product Revenues (REV)				375972	1		$3,158.16	$0.00
By-products & Waste Streams								
By-Product		Benzene	kg	6224	1	$0.65	$32.12	$0.00
By-Product		OX	kg	1321	1	$0.95	$10.02	$0.00
Total Byproducts and Wastes (BP)				7545	2		$42.13	$0.00
Raw Materials								
Feed		Methanol	kg	128168	1	$0.34	$343.49	$0.00
Feed		NAPTHA	kg	419048	1	$0.50	$1,676.19	$0.00
Total Raw Materials (RM)				547,216	2	$0.84	$2,019.68	$0.00
							$1,180.62	**$0.00**

CONSUMABLES

Consumables Types Add Remove	ASPEN Stream	Description	Units	Units/hr hr	Units/Unit MainProduct	Price $/Unit	$MM/yr $MM	$/Unit Main Product
Solvent		DTBB	kg	10.75	1	$20.00	$1.72	$0.00
Solvent		TBB	kg	5.37	1	$20.00	$0.86	$0.00
Total Consumables (CONS)				16.12	2		**$2.58**	**$0.00**

Figure A8-4: Revenues and raw materials calculations from excel.

FIXED OPERATING COSTS

Operating Variabels				Description	$M/yr $M	$/unit main product
Labor						
Operators per Shift Position	4.8					
Number of shift positions	3		$30,000.00	$/yr each	$432.00	$1,149.02
Supervision			25%	of Operating Labor	$108.00	$287.26
Direct Ovhd.			45%	of Labor & Superv.	$243.00	$646.32
Maintenance			3%	of ISBL Investment	$2,354.26	$6,261.80
Overhead Expense						
Plant Overhead			65%	of Labor & Maint.	$2,039.22	$5,423.86
Tax & Insurance			2%	of Fixed Investment	$1,569.51	$4,174.54
Interest on Debt Financing			0%	of Fixed Capital	$0.00	$0.00
Type1			0%	of Working Capital	$0.00	$0.00
Type2						
					$6,745.99	$17,942.81

EXECUTIVE SUMMARY

Final Costs	$MM/yr $MM	$/unit main product
Variable Cost of Production	$2,059.67	$0.00
Fixed Cost of Production	$6.75	$17,942.81
Cash Cost of Production	$2,066.42	$17,942.81
Gross Profit	$1,091.75	-$17,942.81
Total Cost of Production	$4,126.09	$17,942.81

Figure A8-5: Fixed operating costs and final cost calculations from excel.

A9.) –Detailed Equipment Costs

Equipment	ASPEN Block	Purchased Cost	Installed Cost
Distillation Columns		$5.91	$23.63
Tower	T-201	$0.67	$2.69
Sieve Trays		$0.002187	$0.01
Tower	T-202	$2.31	$9.23
Sieve Trays		$0.00	$0.01
Tower	T-203	$1.47	$5.89
Sieve Trays		$0.00	$0.01
Tower	T-204	$0.96	$3.86
Sieve Trays		$0.00	$0.01
Tower	T-205	$0.48	$1.92
Pressure Vessels and Drums		$0.00	$0.00
Horizontal Vessel	D-202	$0.27	$1.10
Horizontal Vessel	D-203	$0.01	$0.05
Horizontal Vessel	D-204	$0.03	$0.12
Horizontal Vessel	D-205	$0.06	$0.24
Horizontal Vessel	D-206	$0.03	$0.13
Horizontal Vessel	D-201	$0.07	$0.30
Heat Transfer Equipment		$0.06	$0.25
Floating Head Shell & Tube	E-205	$3.52	$12.30
Kettle Reboiler	E-206	$0.11	$0.38
Floating Head Shell & Tube	E-207	$0.08	$0.29
Kettle Reboiler	E-208	$0.13	$0.46
Floating Head Shell & Tube	E-209	$0.27	$0.94
Kettle Reboiler	E-210	$0.23	$0.79
Floating Head Shell & Tube	E-211	$0.81	$2.85
Kettle Reboiler	E-212	$0.14	$0.50
Floating Head Shell & Tube	E-213	$0.27	$0.95
Kettle Reboiler	E-214	$0.28	$0.97
Floating Head Shell & Tube	E-201	$0.40	$1.40

Floating Head Shell & Tube	E-202	$0.30	$1.05
Floating Head Shell & Tube	E-203	$0.30	$1.05
Floating Head Shell & Tube	E-204	$0.02	$0.08
Pumps and Drivers		$0.17	$0.60
Single Stage Centrifugal Pump	P-203	$0.21	$0.82
Electric Motor		$0.02	$0.06
Single Stage Centrifugal Pump	P-204	$0.00	$0.00
Electric Motor		$0.01	$0.03
Single Stage Centrifugal Pump	P-205	$0.00	$0.00
Electric Motor		$0.03	$0.11
Single Stage Centrifugal Pump	P-206	$0.00	$0.00
Electric Motor		$0.04	$0.16
Single Stage Centrifugal Pump	P-207	$0.00	$0.00
Electric Motor		$0.04	$0.17
Single Stage Centrifugal Pump	P-208	$0.00	$0.00
Electric Motor		$0.03	$0.11
Single Stage Centrifugal Pump	P-201	$0.00	$0.00
Electric Motor		$0.03	$0.14
Single Stage Centrifugal Pump	P-202	$0.00	$0.00
Electric Motor		$0.02	$0.07
Reactors		$0.11	$0.42
Reactor Vessel	R-201	$0.11	$0.42
Solids Equipment		$0.00	$0.00
TOTAL EQUIPMENT COST		$10.01	$38.28

Figure A9-1: Equipment costs from excel.

A10.) Heat Integration:

The heat integration of the process is investigated to potentially lower capital and operational costs. With the reduction is energy consumption, the CO2 emissions to the environment is reduced. Aspen Energy Analyzer® is used to design a heat network system that minimizes total costs from equipment and operation. Assumptions were made and conventions were used for the process to achieve near target values of cost and energy. Aspen energy analyzer® solves the heat integration problem by solving superstructures to minimize the total annual cost.[10]

Utilities:

Hot and cold utilities are selected from Aspen energy analyzer® utilities data bank after analyzing temperature levels of the involved process streams (such as reboilers, condensers, etc…).

A fire heater utility is used to heat the process streams up to desired operational temperature condition. The high heat generated by the bed reactor is recovered by using the hot reactor effluent to preheat the other streams that require a hot utility. The change in the stream enthalpy is set to equal the heat of reaction. The rest of the utilities are provided by LP and MP steams at temperatures around 200 °C which is enough to provide heating in all reboilers and preheating of feed streams (maximum temperature of operation of columns is about 135°C). Cooling water is used as the sole cold utility to provide cooling up to the lowest temperature of the process streams since condensers operate at ambient condition.

Table A10-1: Utilities for heat integration of toluene methylation

Utility		Cost Index [Cost/s]	Load [kJ/h]	% of Target
CW		6.271e-002	8.874e+008	166.9
MPSTEAM		2.167	7.515e+008	INF
LPSTEAM		7.857e-002	3.316e+007	INF
Fired Heat (1000)		0.0000	0.0000	0.0000
	Total	2.309	---	---

Cost Basis for Heat Integration:

Aspen energy analyzer® generated costs are used in comparison to costs calculated from aspen before heat integration to verify costs reduction. Default cost parameters available through Aspen energy analyzer® cost databank is used to calculate all equipment and operation costs. All the heat exchangers are assumed to be shell and tube heat exchangers for simplicity and capital cost of furnaces are also calculated.

$$CAPEX_{HX} = a + b\left(\frac{A}{N_{shell}}\right)^{c} \times N_{shell} \qquad [10]$$

$$CAPEX_{f} = a + b(Duty)^{c} \qquad [10]$$

Where

$CAPEX_{HX}$[=]installed capital cost of heat exchanger

$CAPEX_{f}$[=]installed capital cost of furnace

a[=] installation cost of a heat exchanger

b, c[=]duty and area related cost parameters

A[=] heat transfer area of a heat exchanger

$Duty$[=]amount of energy being transferred in a heat exchanger

N_{shell}[=]number of shells in heat exchanger

The default values of factors a, b and c are taken from Aspen energy analyzer® and are respectively

1000, 800 and 0.8. $CAPEX_{HX} = 1000 + 800\left(\frac{68727}{147}\right)^{0.8}$ **x 147** = $ 16,079,864

$CAPEX_{HX} = 1000 + 800\left(\frac{68727}{147}\right)^{0.8}$ **x 147** = $ 16,079,864

$CAPEX_f = 1000 + 800(0)^{0.8}$= $1,000 (duty =0 due to heat integration)

$CAPEX =$**$ 16,080,864**

$OPEX = \Sigma(C_{HU} + Q_{HU}) + \Sigma(C_{CU} + Q_{CU})$ [10]

$OPEX$[=]**Oprating cost** , $\$/\mathbf{yr}$

C_{HU}, C_{CU} [=]Cost of hot and cold utilities per unit energy, $\frac{\$}{\mathrm{kJ}}$

Q_{HU}[=]energy target of hot utility, kW

Q_{CU}[=]energy target of cold utility, kW

$OPEX = \$\ 6,649,920$

$TAC = \wedge * CAPEX + OPEX$

TAC[=]Total operational cost

Where $\Lambda = \frac{\left(1+\frac{ROR}{100}\right)^{PL}}{PL}$ [10]

The default values of 10% and 5 years are respectively used for ROR (rate of return) and plant life (PL). [10]

$$TAC = \frac{\left(1+\frac{ROR}{100}\right)^{PL}}{PL} * CAPEX + OPEX$$

$TAC = \frac{\left(1+\frac{10\%}{100}\right)^{5}}{5} * \$\ 16,080,864 + \$\ 6,649,920$, TAC **= $9,882,206/year**

Optimum Minimum Approach temperature (ΔT_{min}):

The $\mathbf{\Delta T_{min}}$ of 15°C gives a total annual cost of $9,882,206/year. Figure # shows the composite curve at $\mathbf{\Delta T_{min}}$ of 15°C. Figure # shows the energy targets for maximum heat energy recovery, the minimum area targets required along with the minimum cost targets.

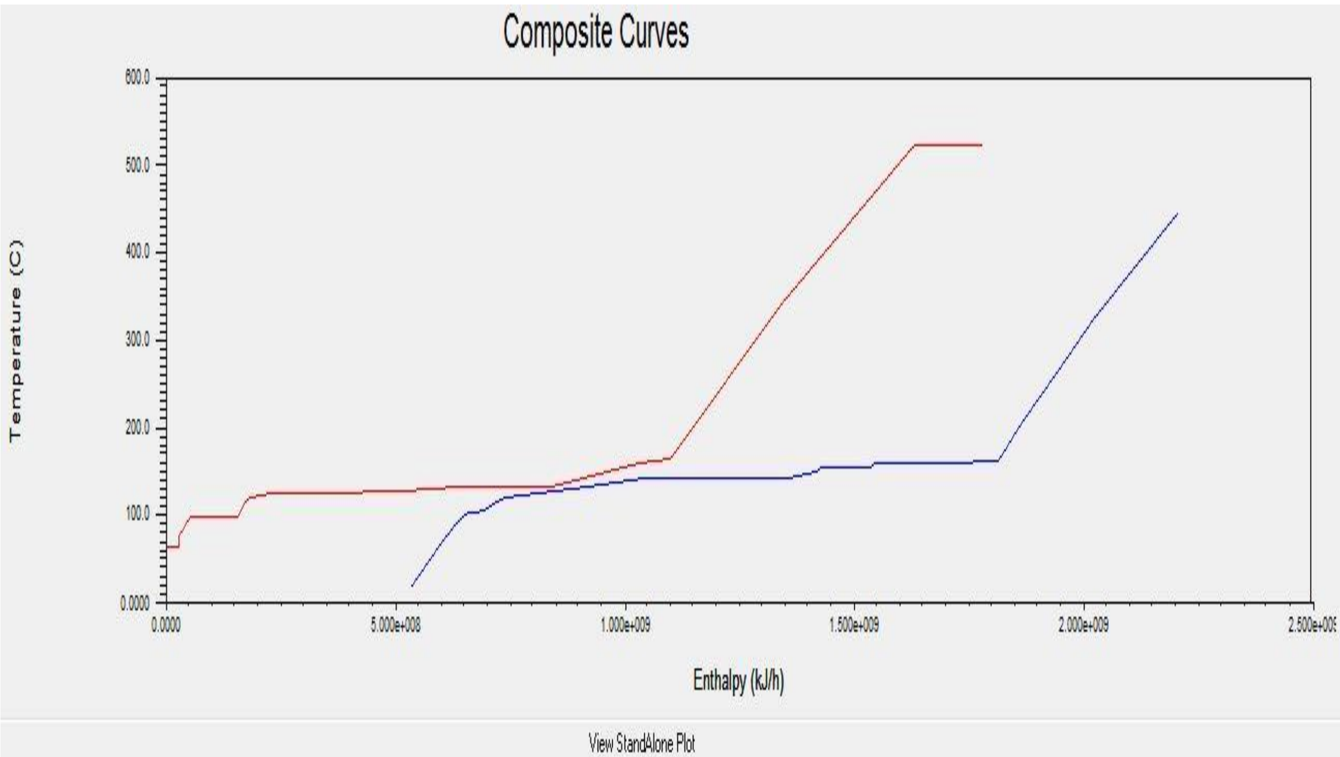


Figure A10-1: Composite curves of heat integration of toluene methylation at ΔT_{min} of 15 °C.

Network Cost Indexes

	Cost Index	% of Target
Heating [Cost/s]	2.246	443.5
Cooling [Cost/s]	6.271e-002	166.9
Operating [Cost/s]	2.309	424.4
Capital [Cost]	1.613e+007	66.12
Total Cost [Cost/s]	2.473	311.9

Network Performance

	HEN	% of Target
Heating [kJ/h]	7.846e+008	182.9
Cooling [kJ/h]	8.874e+008	166.9
Number of Units	14.00	87.50
Number of Shells	145.0	91.19
Total Area [m2]	6.873e+004	176.2

Figure A10-2: Heat integration targets

Heat Network Design:

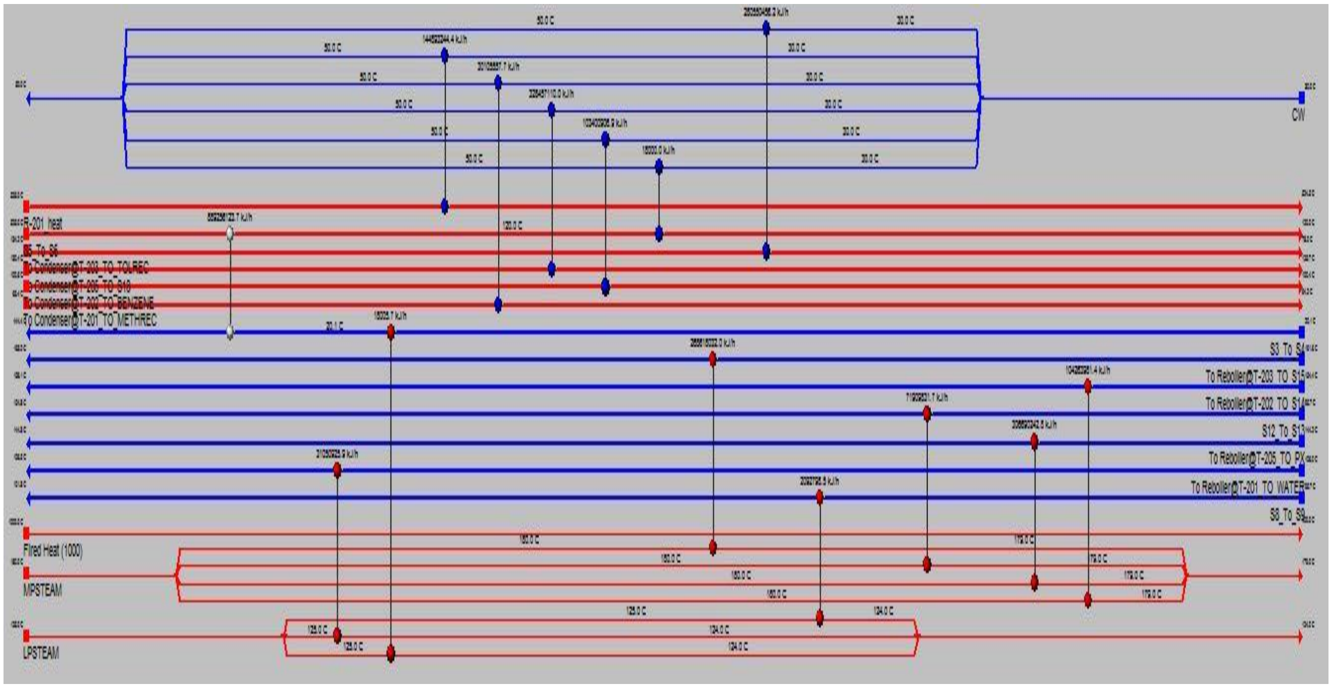


Figure A10-3: Grid diagram of heat network design for the PX process

Table A10-2: Results summary for heat network design

Heat Exchanger	Load [kJ/h]	Cost Index [Cost]	Area [m2]	Shells	LMTD [C]	Overall U [kJ/h.m2.C]	FFactor	Fouling [C.h.m2/kJ]
Reboiler@T-203	268600	431400	1785	4	17.19	8788.8	0.9992	0.0000
Condenser@T-203	280600	161900	592.8	2	64.4	5808.5	0.9523	0.0000
E-203	20930	14120	7.766	1	25.68	10511.2	0.9984	0.0000
E-201_E-202	8.893E+07	1.433E+07	6.19E+04	125	89.93	380.70	0.7946	0.0000
E-204	719100	310200	1255	3	48.57	1198.5	0.9955	0.0000
Reboiler@T-205	306700	215600	865	2	35.03	10133.8	0.9999	0.0000
Reboiler@T-201	31050	39990	92.78	1	18.28	18344.6	0.9998	0.0000

Heat Exchanger	Load [kJ/h]	Cost Index [Cost]	Area [m2]	Shells	LMTD [C]	Overall U [kJ/h.m2.C]	FFactor	Fouling [C.h.m2/kJ]
R-201_heat_Exchanger	144900	113700	437.3	1	484.7	683.50	1.0000	0.0000
Condenser@T-201	30110	64690	196.6	1	23.57	6586.3	0.9928	0.0000
Condenser@T-205	328500	215000	862	2	87.81	4342.5	0.9979	0.0000
Reboiler@T-202	104300	144000	506.5	2	24.25	8499.0	0.9995	0.0000
Condenser@T-202	10340	71190	226.2	1	59.92	7631.4	0.9998	0.0000
E-201_E-202_DummyClrl	18000	10080	5.66E+02	1	79.58	3994.8	1.0000	0.0000

Heat Exchanger	Cold Stream	Cold T in [C]	Tied	Cold T out [c]	Tied	Load [kJ/h]		dT Min Hot [C]
Reboiler@T-203	To Reboiler @T-203_TO_S15	161.6	☼	163	☼	2.69E+08		17.01
Condenser@T-203	CW	20		50		2.81E+08		84.17
E-203	S8_To_S9	95.67	☼	101.8	☼	2.09E+08		23.2
E-201_E-202	S3_To_S4	20.08	☼	444.4	☼	8.89E+08		80.63
E-204	S12_To_S13	95.67	☼	154.8	☼	7.19E+07		25.15
Reboiler@T-205	To Reboiler @T-205_TO_PX	144.2	☼	144.8	☼	3.07E+08		35.24
Reboiler@T-201	To Reboiler @T-201_TO_WATE	106	☼	106.5	☼	3.11E+07		18.54
E-201_E-202_DummyHtr	S3_To_S4	20.06	☼	20.08	☼	1.80E+04		104.9
R-201_heat_Exchanger	CW	30		50		1.45E+08		475
Condenser@T-201	CW	30		50		3.01E+07		15.39

Condenser@T-205	CW	30		50		3.29E+08		80.41
Reboiler@T-202	To Reboiler @T-202_To_S14	154.4	☼	156.1	☼	1.04E+08		23.93
Condenser@T-202	CW	30		50		1.03E+08		50.57
E-201_E-202_DummyClrl	CW	30		50		1.80E+04		70
Heat Exchanger	**Hot Stream**	**Hot T in [C]**	**Tied**	**Hot T out [c]**	**Tied**	**Area [m2]**		**dT Min Cold [C]**
Reboiler@T-203	MPSTEAM	180		179		1784.6		17.38
Condenser@T-203	To Condenser@T-203_TO_TOL	134.2	☼	78	☼	592.83		48
E-203	LPSTEAM	125		124		7.766		28.33
E-201_E-202	S5_To_S6	525	☼	120	☼	61900		99.93
E-204	MPSTEAM	180		179		1255		83.33
Reboiler@T-205	MPSTEAM	180		179		865.03		34.83
Reboiler@T-201	LPSTEAM	125		124		92.777		18.03
E-201_E-202_DummyHtr	LPSTEAM	125		124		4.58E+02		103.9
R-201_heat_Exchanger	R-201_heat	525	☼	524.5	☼	437.35		494.5
Condenser@T-201	To Condenser@T-201_TO_ME	65.39	☼	64.23	☼	196.56		34.23
Condenser@T-205	To Condenser@T-205_TO_S18	130.4	☼	125.7	☼	862.02		95.66
Reboiler@T-202	MPSTEAM	180		179		506.55		24.56

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